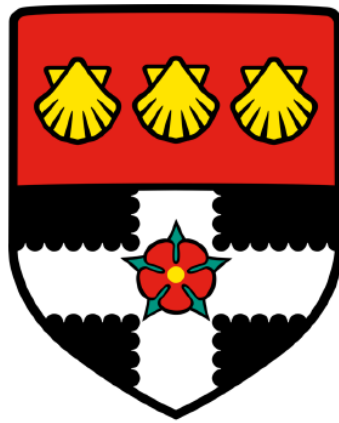


UNIVERSITY OF READING
DEPARTMENT OF METEOROLOGY



Improving Surface Gravity Wave Forecasts Using Data Assimilation

Owen Aljabar

Supervisors: Ivo Pasmans, Sarah Dance, Irina Nikolova and Henrique Coelho

*A thesis submitted in partial fulfillment of the requirements for the degree of
Master of Science in Atmosphere, Ocean and Climate*

August 2024

Declaration of Authorship

I, Owen Aljabar, declare that this thesis titled, “Improving Surface Gravity Wave Forecasts Using Data Assimilation” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:

Date:

Abstract

The ensemble transform Kalman filter (ETKF) is used to assimilate observations of wave height and two measures of period into wave forecasts. The forecasts are produced by operationally by Fugro for a client in the Gulf of Mexico, and the observations come from a buoy belonging to the client in the Southern part of the Gulf. The data assimilation (DA) method used differs from the conventional ETKF method since the lack of access to a forecast model precludes any possibility of cycling. Instead the update step of ETKF is used to correct an entire forecast window at once before loading forecast data from the next window. Observations are assimilated from the first three days of each forecast window (the ‘assimilation window’). One DA experiment (H_s DA) is performed where only wave height is assimilated, and another (all vars DA) is performed where wave height is assimilated alongside the two measures of period.

The background errors are visualised using timeseries of ensemble mean and variance and maps of forecast error covariances. The ensemble variance is seen to be strongly related to the mean, with peaks in variance typically preceding peaks in the mean. The forecast error covariances are seen to have non-zero values at short distances and also sometimes at larger distances. Patterns of wind forcing and travelling waves are offered as physical explanations for these long distance covariances, and the possibility of spurious covariances arising from sampling error is also discussed.

The observation errors are examined using a collocation study and found to be a strong linear function of the observations themselves. Motivated by this, a scaled error approach is used in the DA, where observation errors are set at a constant percentage of the observed value. A minimum error threshold is set for the two measures of period in an attempt to avoid fitting the observations too closely and producing unrealistic increments elsewhere.

Both H_s DA and all vars DA are found to reduce forecast RMSE by around 50% during the assimilation window, and improvements extend up to 0.5 days after the end of the assimilation window in the case of H_s DA, and up to 1.5 days in the case of all vars DA. Both experiments, but especially all vars DA, produce large increments at locations far from the buoy, which are judged to be somewhat unrealistic.

The ensemble spread after DA is examined in comparison to the spread before DA, and to the RMSE. Both DA experiments show a greater overall reduction in spread than in RMSE, thus increasing overconfidence in the ensembles. The errors in forecasts of the maximum significant wave height associated with a cold front on 12/03/2022 are examined, and it is found that at 1-2 days after assimilation, H_s DA improves forecasts of the height of the peak while all vars DA improves forecasts of its timing

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor. . .

Contents

Declaration of Authorship	iii
Abstract	v
Acknowledgements	vii
1 Introduction	1
1.1 The need for Reliable Wave Forecasting	1
1.2 Previous work on DA for Wave Forecasts	2
1.3 Aims of the Project	6
2 Theory	9
2.1 Ocean Waves	9
2.1.1 Physical Processes	9
2.1.2 Linear Wave Theory	10
2.1.3 Wave Spectra	11
2.2 Data Assimilation	13
2.2.1 Vector Representations and Cost Function	13
2.2.2 Kalman Filters	14
2.2.3 Practical Aspects	16
3 Experimental Setup	19
3.1 Model Description and Domain	19
3.2 Background State	21
3.3 ADCP Observations	22
3.4 Twin Experiment	25
3.5 Data Assimilation Setup	25
4 Background and Observation Errors	27
4.1 Background Errors	27
4.2 Diagnosis of Observation Errors	32
5 Results	37
5.1 Increments	37
5.2 RMSE	42
5.3 Correlation	43
5.4 Ensemble Spread	43
5.5 Errors in Maximum H_s	48
6 Conclusions	51
6.1 Summary	51
6.2 Limitations and Future Work	52

A 4DEnVar Method	57
Bibliography	59

List of Figures

1.1	Schematic distributions of maximum wave height in a window, uncorrected (solid) and corrected for forecast bias and uncertainty (dashed)	2
1.2	RMSE in wave height (left) and period (right) from two different observation locations. Figure 30 from Caires, Kim, and Groeneweg, 2018.	3
1.3	RMSE from spectral DA and wave height (H_s) DA, normalised by the RMSE without DA. Results shown for significant wave height H_s , peak period T_p , mean period T_m , directional spread σ_θ , peak direction D_p and mean direction D_m . Figure 3 from Houghton et al. (2022).	4
1.4	Forecast bias and RMSE from altimeter observations (top) and buoy observations (bottom). Figure 6 from Houghton et al., 2023.	5
1.5	An example H_s increment from both LETKF and OI methods. Colours show value of increment, markers show the locations of observation buoys. Figure 3 from Houghton et al., 2023.	5
2.1	Schematic of the generation of wind and swell waves.	10
2.2	An example spectrum from a point in the gulf of Mexico. Calculated from WW3 spectral forecast data.	12
2.3	Left: an ensemble with appropriate spread. Right: an overconfident ensemble. Truth is shown in black, ensemble mean and spread is shown in blue and ensemble members are shown in green.	17
3.1	Model domains shown on map.	20
3.2	Schematic of the forecasting setup used by Fugro. A new 7-day 31-member ensemble forecast is produced at 12Z each day.	21
3.3	Left: H_s (colours) and wave direction (arrows) averaged between 12Z 06/03/2022 and 12Z 12/03/2022. Right: The same averaged between 12Z 13/03/2022 and 12Z 19/03/2022	21
3.4	Left: H_s (colours) and direction (arrows) in the GoM at 18Z on 12/03/2022. Right: Surface analysis chart valid for 18Z on 12/03/2022, taken from NOAA WPC surface analysis archive	22
3.5	Left: T_{m01} at 18Z on 12/03/2022. Right: T_{m01} averaged over the period from 12Z 10/03/2022 to 12Z 12/03/2022	22
3.6	Observed timeseries of significant wave height from ADCP buoy	23
3.7	Timeseries of observed H_s (top) and T_{m01} (bottom) over a three-month period from 01/12/2021 to 01/03/2022	24
3.8	Density plot of H_s (radial axis) and wave direction at the buoy location.	24
3.9	Schematic of the forecast and analysis steps from the ETKF scheme employed in this project.	25
4.1	Timeseries of forecast H_s ensemble mean and variance at the buoy location from the 10/03/2022 model run.	28
4.2	Timeseries of forecast T_{m01} ensemble mean and variance at the buoy location from the 10/03/2022 model run.	29

4.3	Colours showing spatial covariance of H_s at 12Z 12/03/2022. Arrows showing wind averaged over prior 24 hours.	29
4.4	Colours showing spatial covariance of H_s at 12Z 11/03/2022. Arrows showing wind averaged over prior 24 hours.	30
4.5	Colours showing covariance of H_s at buoy at 18Z 13/03/2022 with H_s elsewhere at 18Z 12/03/2022. Arrows showing wave direction at 18Z 12/03/2022	31
4.6	Cross-covariance between H_s at the buoy location and T_{m01} elsewhere at 12Z 12/03/2022. Arrows show wave direction (left) and wind (right)	31
4.7	Observation - background histogram for significant wave height.	33
4.8	A plot of H_s error standard deviation against H_s	34
4.9	Plots of T_{m01} and T_{m02} error standard deviation against T_{m01} and T_{m02} respectively.	34
4.10	Observation minus background histograms for T_{m01} (left) and T_{m02} (right).	35
5.1	Timeseries of H_s (top) and T_{m01} (bottom) at buoy location from 12Z 09/03/2022 model run. No DA ensemble mean is shown in red, H_s DA ensemble is shown in blue, all vars DA ensemble mean is shown in green and truth is shown in black. Blue dotted line indicates the end of the assimilation window.	38
5.2	Timeseries of H_s (top) and T_{m01} (bottom) at 30°N 80° W from 12Z 09/03/2022 model run. No DA ensemble mean is shown in red, H_s DA ensemble is shown in blue, all vars DA ensemble mean is shown in green and truth is shown in black. Blue dotted line indicates the end of the assimilation window.	39
5.3	H_s ensemble mean increments at 12Z 12/03/2022 from all vars DA (top left) and H_s DA (top right), and H_s innovations from the same time (bottom). Taken from 12Z 10/03/2022 forecast run. The buoy location is marked with a cross.	40
5.4	T_{m01} ensemble mean increments at 12Z 12/03/2022 from all vars DA (top left) and H_s DA (top right), and H_s innovations from the same time (bottom). Taken from 12Z 10/03/2022 forecast run. The buoy location is marked with a cross.	41
5.5	RMSE in H_s (left) and T_{m01} (right) ensemble mean at the buoy location as a function of leadtime. RMSE is calculated over a 24hr period. Results shown for no DA (solid line), H_s DA (dashed line) and all vars DA (black dotted line). Blue dotted line indicates the end of the assimilation window.	42
5.6	Maps of RMSE in H_s ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and H_s DA (right). The buoy location is marked with a cross.	44
5.7	Maps of RMSE in T_{m01} ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and H_s DA (right). The buoy location is marked with a cross.	44

5.8	Timeseries of ensemble standard deviation in H_s (top) and T_{m01} (bottom) at the buoy location. Taken from 12Z 06/03/2022 (left) and 12Z 10/03/2022 (right) model runs.	45
5.9	Ensemble standard deviation in H_s at 12Z 11/03/2022, taken from 12Z 06/03/2022 (top) and 12Z 10/03/2022 (bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation reduction from all vars DA (left) and H_s DA (right).	46
5.10	Ensemble standard deviation in T_{m01} at 12Z 11/03/2022, taken from 12Z 06/03/2022 (top) and 12Z 10/03/2022 (bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation reduction from all vars DA (left) and H_s DA (right).	47
5.11	Left: Scatter plot of H_s RMSE against H_s ensemble standard deviation at the buoy location. All vars DA shown with triangles, H_s DA shown with circles and leadtime shown by colour. Lines of best fit shown for all vars DA (dotted) and H_s DA (dashed) as well as 1:1 line (solid). Right: the same for RMSE and ensemble spread in T_{m01}	48
5.12	Scatter plots of timing and magnitude of maximum H_s at buoy location from period 12Z 12/03/2022 to 12Z 13/03/2022. Generated from model run starting 12Z 10/03/2022 (left) and 12Z 08/03/2022 (right). Truth is shown in black, and the no DA, H_s DA and all vars DA ensemble members are shown in red, blue and green respectively.	49
6.1	Locations of buoys with observations of relevant wave variables. The buoy used in this project is marked in black, and other buoys are marked in blue	53
6.2	Schematics of potential DA methods which incorporate cycling. Top: Observations assimilated hourly using a conventional ETKF filter method. Bottom: Observations assimilated every 6 hours using the 4DEnVar smoother method.	54

List of Abbreviations

DA	Data Assimilation
OI	Optimal Interpolation
EnKF	Ensemble Kalman Filter
ETKF	Ensemble Transform Kalman Filter
LETKF	Local Ensemble Transform Kalman Filter
WW3	WAVEWATCH III
GoM	Gulf of Mexico

Chapter 1

Introduction

1.1 The need for Reliable Wave Forecasting

Offshore operations form an important part of many industries such as energy and research, and contribute to the economies of coastal countries round the world. Ocean waves pose a significant risk to offshore operations as they can lead to the loss of life and damage to equipment. This means that accurate wave predictions are a critical safety issue for offshore operations, as offshore operations can only be carried out safely when the waves are sufficiently calm. To provide these predictions the MetOcean Forecasting division of Fugro produces wave forecasts for a client undertaking offshore operations in the Gulf of Mexico. It is important to understand the risk posed by waves in this region since increased cyclone intensity is leading to increased extreme wave heights in the Gulf of Mexico (Appendini et al., 2014). To avoid unnecessary downtime, they produce ensemble forecasts and apply corrections using a proprietary method based on the alpha factor method described in Wu and Gao, 2021.

This proprietary method is referred to by Fugro as a site-specific alpha factor and seeks to correct for the bias and uncertainty present in the forecast by defining the alpha factor as follows:

$$\alpha = \frac{H_{\max}}{H_{\max,\text{cor}}} \quad (1.1)$$

where $H_{\max}$ is the maximum wave height with a 10^{-4} probability of exceedance during a given window. $H_{\max,\text{cor}}$ is the maximum wave height with the same exceedance probability during the same window, after the forecast is corrected for bias and uncertainty. The Rayleigh distribution is used to model the probability distribution of maximum wave height over a window as follows: (cite)

$$\mathbb{P}(H \leq H_{\max}) = \left[1 - e^{\left(-2\frac{H^2}{H_s^2}\right)} \right]^{\frac{t}{T_p}} \quad (1.2)$$

Where H_s is the significant wave height during the window, t is the length of the window and T_p is the mean period during the window. $H_{\max}$ is then chosen so that $1 - \mathbb{P}(H \leq H_{\max}) = 10^{-4}$. $H_{\max,\text{cor}}$ is found by integrating the joint probability density function of H and H_s :

$$\mathbb{P}(H, H_s) = \mathbb{P}(H|H_s)\mathbb{P}(H_s) \quad (1.3)$$

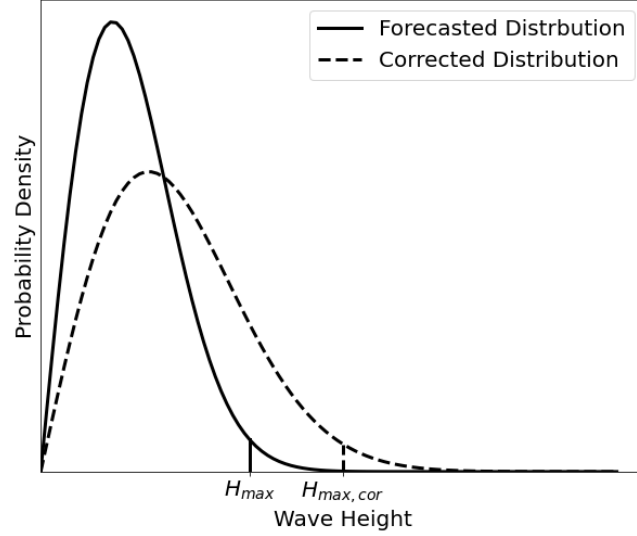


FIGURE 1.1: Schematic distributions of maximum wave height in a window, uncorrected (solid) and corrected for forecast bias and uncertainty (dashed)

Here $\mathbb{P}(H|H_s)$ is the Rayleigh distribution from equation 1.2 and $\mathbb{P}(H_s)$ is assumed to be a normal distribution with mean equal to the bias corrected forecast value of H_s and standard deviation equal to the forecast standard deviation. The bias and standard deviation were calculated from the timeseries of model output and observations spanning from September 2020 to May 2021. Integrating $\mathbb{P}(H, H_s)$ over all values of H_s gives a corrected distribution of maximum wave height during the window which can be used to define $H_{\max, \text{cor}}$ in the same way as $H_{\max}$.

Once the value of α is found, it is used to scale the H_s model output via $H_{s, \text{cor}} = \frac{H_s}{\alpha}$. So if the model predicts $H_s = 1.5\text{m}$ and $\alpha = 0.75$, the scaled forecast value is $H_{s, \text{cor}} = 2\text{m}$. From speaking to Fugro engineers we learned that this method is overly cautious and produces unrealistically high waves when H_s is large.

Qualitatively speaking, α will always be less than 1, since forecast uncertainty acts to broaden the distribution of maximum wave height. This means that $H_{\max, \text{cor}}$ is always more than $H_{\max}$, as illustrated in figure 1.1.

As well as these forecasts, Fugro has a buoy based near to the client which produces hourly observations of the sea state. Data assimilation refers to a range of techniques used to integrate observations into a forecast, with the aim of making the forecast more accurate. In this project, we explore the potential of DA to reduce uncertainty and error in these wave forecasts. We hope that DA will give better results than the alpha factor method currently in use.

1.2 Previous work on DA for Wave Forecasts

Several studies have been completed previously investigating the effectiveness of DA methods to improve wave forecasts. In this section, some of these studies and their findings are summarised, and the differences between this study and previous ones are discussed.

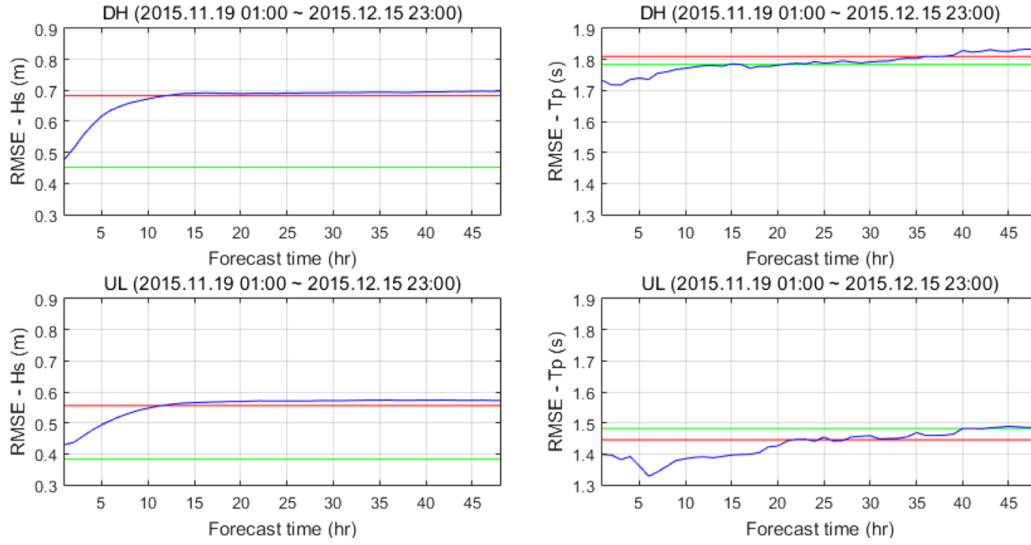


FIGURE 1.2: RMSE in wave height (left) and period (right) from two different observation locations. Figure 30 from Caires, Kim, and Groeneweg, 2018.

Caires, Kim, and Groeneweg, 2018 investigated the effect on wave forecasts of using an ensemble Kalman filter (EnKF) method to adjust the wind fields used as forcing in a wave model. They used hourly observations from six locations. Their results are shown in figure 1.2.

An improvement is seen in forecasts of wave height and period, but it is relatively small compared to other studies and short lived, not extending past 24hrs in any instance (check). This may be due to the indirect relationship between the assimilated wind data and the forecasted wave variables. We hope that assimilating the forecasted variables directly will lead to improved performance over this wind-based approach.

Another possible reason for the relatively poor performance in this case is the small number of observations used. Observations were available hourly from only six different locations, while operational DA systems commonly assimilate upwards of 10^6 observations at a time. This study is also limited in this regard, as data is only available from on buoy. A study which used many more observations is Houghton et al., 2022, in which an optimal interpolation (OI) method (cite) was used to compare the effects of assimilating wave height and spectra. In this study, observations came from the global Sofar spotter buoy network, which contains over 600 buoys. A plot of their results is shown in figure 1.3.

Figure 1.3 shows that both variables give very similar improvements in predictions of wave height, while spectral assimilation gives better results for other quantities such as period and direction. They also found that the spectral DA did particularly well in swell-dominated fields when compared to assimilating wave height. This shows that the type of observations assimilated can have a large effect on the performance of a DA system.

In this project we assimilate two measures of period, T_{m01} and T_{m02} . These variables are related to different moments of the spectrum (see equations 2.8 - 2.11), so we hope that assimilating both T_{m01} and T_{m02} as well as wave height will have some of the benefits of spectral DA identified in Houghton et al., 2022. Since assimilating a full wave spectrum involves much more data than bulk variables such as wave height

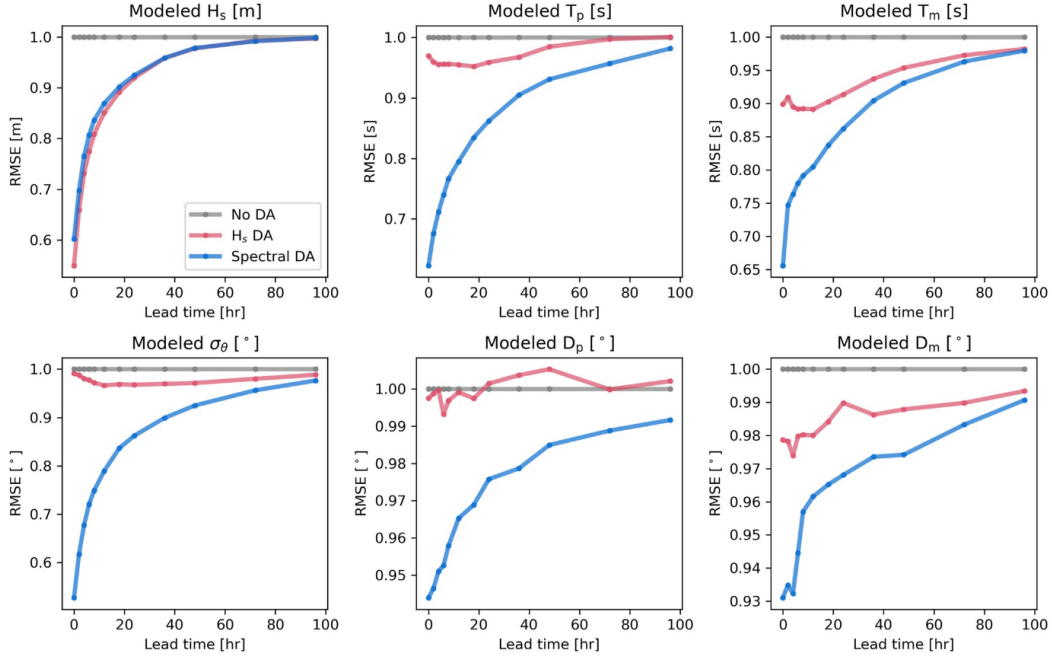


FIGURE 1.3: RMSE from spectral DA and wave height (H_s) DA, normalised by the RMSE without DA. Results shown for significant wave height H_s , peak period T_p , mean period T_m , directional spread σ_θ , peak direction D_p and mean direction D_m . Figure 3 from Houghton et al. (2022).

and period, we hope also that our method will incur less computational cost than assimilating a full spectrum.

To investigate the effect of the DA method used, Houghton et al., 2023 assimilated wave height observations from satellite altimeters and Sofar spotter buoys. They compared two different DA methods, OI and the ensemble-based local ensemble transform Kalman filter (LETKF) (Hunt, Kostelich, and Szunyogh, 2007). Plots of their results are included in figure 1.4.

They found that both methods reduced forecast error and bias out to 2.5 days lead-time, with LETKF having slightly greater forecast skill overall. In addition to this, they found that the LETKF gave more physically realistic increments due to its flow dependent background error covariance matrix. A plot showing increments from each method is shown in figure 1.3. This shows that the as well as the observations, the DA method used affects DA performance. In this project we use the ensemble transform Kalman filter (ETKF) method, which is similar to the LETKF except it does not use localisation, which allows observations to impact the state over long distances. This method was chosen over LETKF partly because it leads to a simpler DA implementation, and partly because some long-distance covariances were found in the forecast data which appear to be genuine (section 4.1).

In this project, we apply DA to a specific forecasting context and client in the Gulf of Mexico, and so we are constrained by the observation data available in this context. As seen above, the studies with more observations saw better DA performance, and our results should be evaluated with this limitation in mind. Another limitation which none of the previous studies were subject to is that we are unable to run the wave model ourselves, which is a central step of most DA systems. We hope that the

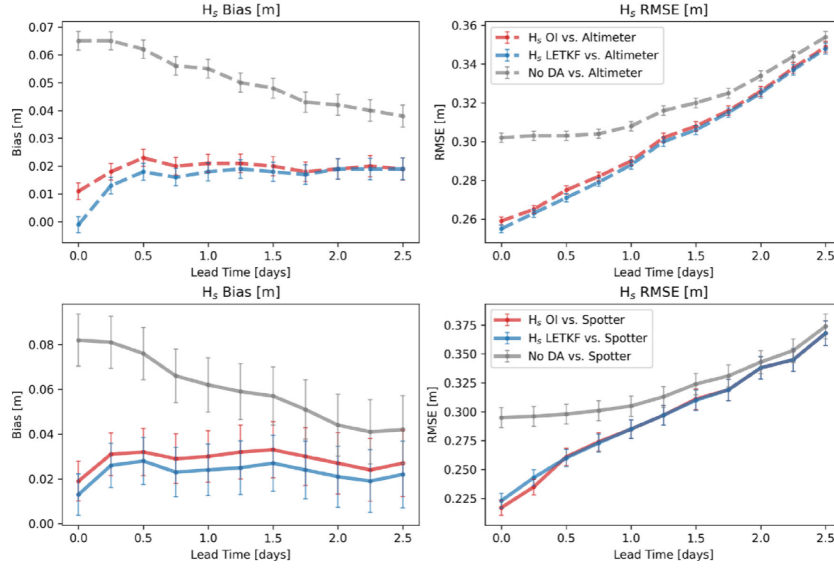


FIGURE 1.4: Forecast bias and RMSE from altimeter observations (top) and buoy observations (bottom). Figure 6 from Houghton et al., 2023.

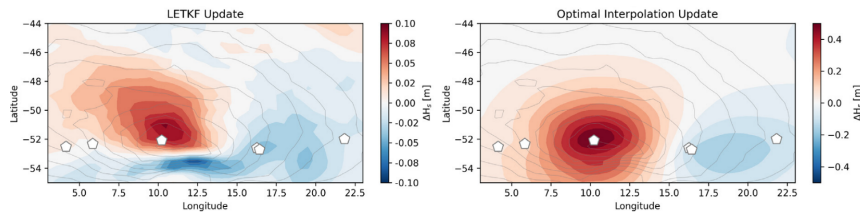


FIGURE 1.5: An example H_s increment from both LETKF and OI methods. Colours show value of increment, markers show the locations of observation buoys. Figure 3 from Houghton et al., 2023.

knowledge gained from these previous studies will allow us to get good results in spite of these limitations.

1.3 Aims of the Project

The main research question in this project is “**What kind of reduction in forecast error can be achieved by assimilating H_s , T_{m01} and T_{m02} using the ensemble transform Kalman filter method?**”

To this end, we perform a twin experiment using the ensemble transform Kalman filter (ETKF) to assimilate observations of the three variables. A twin experiment is a DA experiment in which fictional observations are generated and assimilated in place of real observations (see section 3.4 for details). We perform one experiment in which all three aforementioned variables are assimilated, and one experiment in which wave height alone is assimilated. The reasoning for this is that wave height extremes are of particular interest to Fugro’s client due to their impact on operational safety, so we consider H_s to be the most important variable of the three. We examine the results of these two experiments in comparison with each other, and with the forecast data before DA, to evaluate the performance of DA in both instances.

Various aspects of the results will be examined to gain a full understanding of the DA performance. These are:

- Increments. The changes made to the forecast by the DA system are called increments, and they can aid understanding of how DA is modifying the model state. If the DA system is performing well, we expect to see physically realistic increments.
- The root mean square error (RMSE) of the ensemble mean. We expect the DA to move the ensemble mean closer to the true state so RMSE will be used to quantify and test this. The dependence of RMSE on forecast leadtime and location within the domain will be examined.
- The correlation between ensemble mean and the true state. DA should make the ensemble mean more closely correlated with the true state, so we will test this hypothesis.
- Ensemble spread. This is an important measure of uncertainty in ensemble forecasts, which can give a good estimate of forecast error and allow a forecast to inform decisions with greater or lesser confidence. We hope that DA will reduce the ensemble spread, thus allowing decisions to be made with greater confidence based on the forecasts. The ensemble spread is compared to the RMSE to check for ‘overconfidence’ in the post-DA ensemble (see section 2.2.3).
- Errors in maximum H_s . Extremes of H_s are of particular interest in these forecasts as these are the events which cause danger and require the pausing of offshore operations. We will examine the effectiveness of DA to reduce error in the timings and magnitudes of wave height maxima.

T_{m02}

Chapter 2

Theory

This chapter outlines the theoretical aspects of this project, covering both the linear wave theory used in the wave model and the theory behind data assimilation methods.

2.1 Ocean Waves

Ocean waves are a distinctive visual feature of the ocean surface, and very important to offshore operations such as wind farms. This section will describe some of the physical processes influencing ocean waves, as well as the theory of linear waves and wave energy spectra as modelled by the WAVEWATCH III (WW3) model used in this project (WW3DG, 2019).

2.1.1 Physical Processes

Ocean waves are an instance of surface gravity waves, meaning they propagate on the interface between two fluids of different densities. Their restoring force is gravity. The focus of this project is on waves with periods from 10s - 0.25s, which are generated by wind, although ocean waves can exist on a wide range of scales and be influenced by other factors such as surface tension and local air pressure variations (Holthuijsen, 2007, pp. 3-7).

There are multiple hypothesised physical mechanisms for the generation of waves by wind. Phillips (1957) shows that small-scale turbulent eddies induce small surface waves at frequencies matching the frequency of the turbulence. Miles, 1957 describes how the surface height variations from these small waves lead to energy transfer from the wind to the waves, thus increasing wave height. A more detailed overall description of the various processes is given in Janssen (2004), chapter 3.

The wind almost always acts as a source of wave energy, so there has to be a process present to dissipate waves and act as an energy sink. These processes are poorly understood but Janssen (2004), section 4.9 describes multiple physical mechanisms which may contribute to wave dissipation. One important process affecting dissipation is wave breaking, which occurs in near shorelines when the wave height gets too large compared to the water depth. It can also occur when wave height gets too large compared to wavelength. This steepness induced breaking is also known as white capping. Water's molecular viscosity can lead to the damping of shorter waves and larger waves can be damped by the surface eddies induced by wave breaking.

Ocean waves in deep water are dispersive, and waves with longer wavelengths travel faster than waves with shorter wavelengths (equations 2.3-2.5). This leads to a phenomenon known as swell, which is the name given to long waves that have travelled

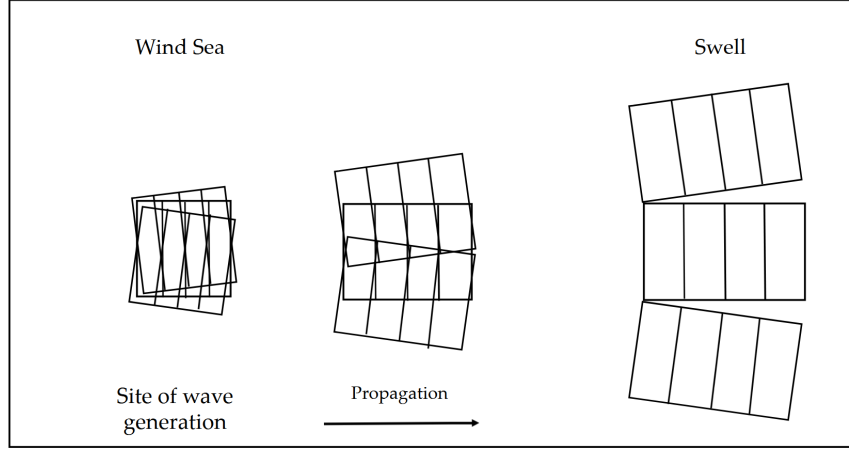


FIGURE 2.1: Schematic of the generation of wind and swell waves.

far from their source. Due to dispersion these waves have no short-wavelength component, so swell waves have longer wavelengths and periods than waves near the site of generation (known as wind sea). The other feature of swell waves is their regular appearance. While a wind sea has wave components in many different directions, by the time the waves have travelled away from the source these different components will spread out so that they do not overlap, and the resulting swell will be almost unidirectional. See figure 2.1 for a schematic.

2.1.2 Linear Wave Theory

This section will outline the linear wave theory relevant to this project, as derived in Vallis (2017), page 253. Linear wave theory is an approximation, and other higher-order theories of surface gravity waves exist, such as Stokes waves (cite) and cnoidal waves (cite). These higher-order theories will not be considered here since the WAVEWATCH III model used in this project models the ocean surface as a linear superposition of simple linear waves.

Parameters describing linear waves include the wavenumber k , direction θ and frequency ω . The first two describe the wave vector $\mathbf{k}$, which has magnitude k and direction θ i.e.

$$\mathbf{k} = k \cos \theta \hat{\mathbf{i}} + k \sin \theta \hat{\mathbf{j}} \quad (2.1)$$

where $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ are unit vectors pointing East and North respectively. This means that θ is the angle between $\mathbf{k}$ and $\hat{\mathbf{i}}$, measured anticlockwise from $\hat{\mathbf{i}}$. ω here denotes the absolute frequency measured in a fixed frame of reference. If the medium is subject to currents, this will differ from the relative frequency σ measured in a frame of reference moving with the mean current, due to the Doppler effect. This difference is quantified by the equation

$$\omega = \sigma + \mathbf{k} \cdot \mathbf{U} \quad (2.2)$$

and σ and k are related by the dispersion relation

$$\sigma^2 = gk \tanh kD \quad (2.3)$$

The dispersion relation has two limiting forms:

- Deep water waves, where $kD \gg 1$, so $\tanh kD \approx 1$ and $\sigma^2 \approx gk$.
- Shallow water waves, where $kD \ll 1$, so $\tanh kD \approx kD$ and $\sigma^2 \approx gk^2D$

The phase speed is then given by

$$c = \frac{\sigma}{k} \approx \begin{cases} \sqrt{\frac{g}{k}} & \text{in deep water} \\ \sqrt{gD} & \text{in shallow water} \end{cases} \quad (2.4)$$

and the group speed may be expressed as

$$c_g = \frac{\partial \sigma}{\partial k} = \frac{1}{2}c \left(1 + \frac{2kD}{\sinh 2kD} \right) \approx \begin{cases} \frac{1}{2}c & \text{in deep water} \\ c & \text{in shallow water} \end{cases} \quad (2.5)$$

The phase speed is the speed at which the wave peaks and troughs move, whereas the group speed is the speed at which wave groups move. The group speed is then the speed at which energy is transported by the wave. The wave travels in the direction of the wave vector $\mathbf{k}$ so the group and phase velocities are given by

$$\begin{aligned} \mathbf{c} &= c \cos \theta \hat{\mathbf{i}} + c \sin \theta \hat{\mathbf{j}} \\ \mathbf{c}_g &= c_g \cos \theta \hat{\mathbf{i}} + c_g \sin \theta \hat{\mathbf{j}} \end{aligned}$$

We can now use scale analysis to see that ω and σ generally have similar values. The ratio between $\mathbf{k} \cdot \mathbf{U}$ and σ in equation 2.2 is

$$\frac{\mathbf{k} \cdot \mathbf{U}}{\sigma} \sim \frac{kU}{\sigma} = \frac{U}{c}$$

Where $U = |\mathbf{U}|$ is the mean current speed. Since $c \gg U$ in general, this ratio is small and so $\omega \approx \sigma$.

2.1.3 Wave Spectra

Note that equations 2.2 - 2.5 are valid for linear waves of a single frequency and direction. Real ocean waves are more complicated than this, containing multiple frequencies propagating in multiple directions. The WAVEWATCH III model used in this project models ocean waves as a superposition of these simple linear waves.

Surface waves with multiple frequencies and directions are represented using a wave spectrum $F = F(\sigma, \theta)$, which is also a function of position and time. F represents the sea surface height variance associated with a particular frequency σ and direction θ , so the total height variance is

$$\overline{\eta^2} = \iint F d\sigma d\theta \quad (2.6)$$

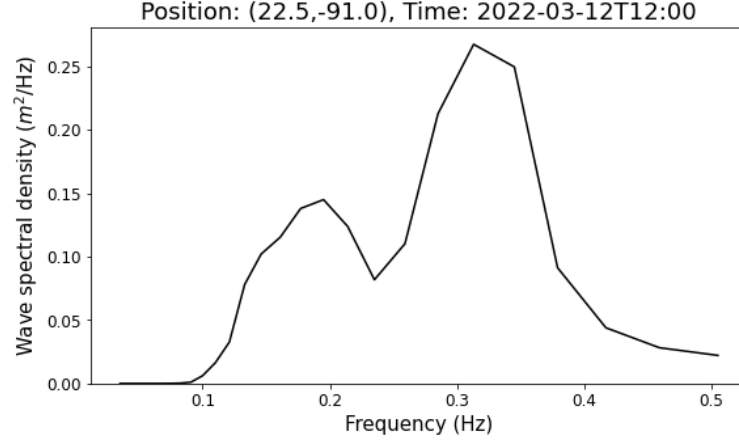


FIGURE 2.2: An example spectrum from a point in the gulf of Mexico. Calculated from WW3 spectral forecast data.

This is related to the wave energy via $E = \rho g \overline{\eta^2}$. A rigorous definition of F is given in Holthuijsen (2007), section 3.5.6. One can also define the direction-integrated spectrum $S(\sigma) = \int F(\sigma, \theta) d\theta$ which quantifies the surface height variance as a function of frequency. This can be integrated to calculate the energy contained in a specific frequency band $\Delta\sigma$:

$$\Delta E = \rho g \int_{\Delta\sigma} S(\sigma) d\sigma \quad (2.7)$$

S is a helpful diagnostic from which information about the appearance of the waves can be inferred. A widely spread spectrum implies irregular waves with many different frequency components whereas a tight spectrum implies regular waves with one dominant frequency. Low frequency waves correspond to swell, while high frequency waves correspond to a wind sea. An example of a direction-integrated spectrum is plotted in figure 2.2. This spectrum comes from a point in the GoM that is far from any shorelines. We can see two peaks on the spectrum, one at about 0.3 Hz which corresponds to wind waves and one at about 0.2 Hz, which corresponds to swell.

Here, we also define the wave spectral moments which are derived from the direction-integrated spectrum:

$$m_k = \int S(\sigma) \sigma^k d\sigma, \quad m = \dots, -1, 0, 1, \dots \quad (2.8)$$

Note that $m_0 = \overline{\eta^2}$ is the overall surface height variance. These moments can be used to derive other important wave variables from the spectrum, such as significant wave height and wave period. In this project we focus on the significant wave height H_s and two measures of frequency T_{m01} , T_{m02} . These variables are defined below:

$$H_s = 4\sqrt{m_0} \quad (2.9)$$

$$T_{m01} = \frac{m_0}{m_1} \quad (2.10)$$

$$T_{m02} = \sqrt{\frac{m_0}{m_2}} \quad (2.11)$$

Finally we define the wave action

$$N(\sigma, \theta) = \frac{F(\sigma, \theta)}{\sigma}$$

N is conserved in the absence of sources and sinks (Bretherton and Garrett, 1968), so it is very useful for wave modelling. WW3 works by evolving the wave action according to the effects of sources and sinks (see section 3.1).

2.2 Data Assimilation

Data assimilation (DA) refers to a group of statistical methods for incorporating observations into forecasts. It requires a model state and some observations, and the assumption that the observations are a function of some correct underlying state of the system. The goal is to find the state that is most likely given the forecast and observation data, assuming some distributions for the errors in each.

2.2.1 Vector Representations and Cost Function

We take a vector $\mathbf{x} \in \mathbb{R}^n$ to represent the state of the system, assuming it has a normal background distribution $\mathbf{x} \sim N(\mathbf{x}_b, \mathbf{B})$. We take a vector $\mathbf{y} \in \mathbb{R}^p$ to represent a set of observations, and assume $\mathbf{y}$ is related to $\mathbf{x}$ via an observation operator $\mathcal{H} : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and an error term ε :

$$\mathbf{y} = \mathcal{H}(\mathbf{x}_{\text{true}}) + \varepsilon$$

where $\mathbf{x}_{\text{true}}$ is the unknown true state of the atmosphere. The error term ε is assumed to be normally distributed with mean zero and covariance $\mathbf{R}$: $\varepsilon \sim N(0, \mathbf{R})$. We call the vector spaces containing $\mathbf{x}$ and $\mathbf{y}$ the state space and observation space respectively. In most operational contexts, the dimensions of these spaces are very large: $\mathbf{x}$ has one component for each variable involved in the assimilation at each model gridpoint, so typically $n \sim 10^8$. $\mathbf{y}$ has one component for each observed variable at each observation location, so $p \sim 10^6$. This means that considerable computational resources are required in most cases and computational efficiency is a key consideration in any operational DA implementation.

In our case, we have three variables on a 181×210 grid, so the dimension of our state space is a relatively modest 114030. We have only one observation location measuring the same three variables, so the dimension of the observation space here is only 3.

In our case $\mathcal{H}$ is linear i.e. there is a matrix $\mathbf{H} \in \mathbb{R}^{n \times p}$ such that $\mathcal{H}(\mathbf{x}) = \mathbf{H}\mathbf{x}$, and so $\mathbb{P}(\mathbf{y}|\mathbf{x}) = N(\mathbf{H}\mathbf{x}, \mathbf{R})$. We can then express the distribution of $\mathbf{x}$ given some set of observations $\mathbf{y}$ using Bayes' theorem:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = \mathbb{P}(\mathbf{y}|\mathbf{x}) \frac{\mathbb{P}(\mathbf{x})}{\mathbb{P}(\mathbf{y})} \quad (2.12)$$

This allows $\mathbb{P}(\mathbf{x}|\mathbf{y})$ to be determined since $\mathbb{P}(\mathbf{y}|\mathbf{x})$ and $\mathbb{P}(\mathbf{x})$ are known, and $\mathbb{P}(\mathbf{y})$ is independent of $\mathbf{x}$ so it acts as a normalisation constant. Since all the distributions involved are normal, $\mathbb{P}(\mathbf{x}|\mathbf{y})$ will be normally distributed also:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = A(\mathbf{y}) e^{-\frac{1}{2}(\|\mathbf{x}-\mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \|\mathbf{y}-\mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2)} \quad (2.13)$$

adopting the notation $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{a}^\top \mathbf{M} \mathbf{a}$. $A(\mathbf{y})$ in this equation is a normalising constant whose value is chosen so that the integral of $\mathbb{P}(\mathbf{x}|\mathbf{y})$ over all values of $\mathbf{x}$ is equal to 1 for each $\mathbf{y}$. Finding the most likely value of $\mathbf{x}$ given this distribution is equivalent to minimising a cost function

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2 \quad (2.14)$$

The first term of 2.14 represents the deviation of $\mathbf{x}$ from the background state and the second term represents the deviation from the observations, with the overall cost function being a sum of these two weighted by the background and observation error matrices. This means that when the background error is large compared to the observational error, the observations will be given more weight and vice versa.

2.2.2 Kalman Filters

The state that minimises the cost function in equation 2.14 is called the analysis $\mathbf{x}^a$ and may be expressed as

$$\mathbf{x}^a = \mathbf{x}_b + \mathbf{K} \mathbf{d}_b^o \quad (2.15)$$

where $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}$ is the innovation and $\mathbf{K} = \mathbf{B}\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1}$ is the Kalman gain (derived in Asch, Bocquet, and Nodet, 2016, pp. 92-96). Note that the increment $\mathbf{d}_b^a = \mathbf{x}^a - \mathbf{x}_b$ is given by $\mathbf{B} \left(\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{d}_b^o \right)$, so it always lies in the column space of $\mathbf{B}$. This means that $\mathbf{B}$ is crucial to determining how information from the observations is spread around the domain by the DA system. In particular, if $\mathbf{B}$ is rank deficient, then all possible increments lie within a strict subspace of the state space. This reduces the degrees of freedom in the corrections the DA system can make, which can limit the performance of the system.

One class of DA methods which use equation 2.15 are called Kalman filters, in which context it is called the analysis step. Kalman filters alternate this analysis step with a forecast step in which a forecast model is used to evolve the state forward to the next observation time. Rather than a fixed background error matrix $\mathbf{B}$, Kalman filters use a flow-dependent error covariance matrix $\mathbf{P}$ (equations 2.19 and 2.20) which is explicitly part of the DA output and which is evolved forward in time by the model along with the state.

In this project we borrow the Kalman filter analysis step to update the model state. Our method is not strictly speaking a Kalman filter since we do not have access to the forecast model, which means we are unable to perform the forecast step required

in a Kalman filter method. This process of using the forecast model to evolve the DA corrections forward to the next assimilation time is called cycling, and is a central part of most DA systems. We view the fact that we are unable to do this as an important limitation of this study.

To get around the lack of cycling, we instead make use of a single analysis step to update an entire 7-day forecast window at once. This requires augmenting the state vector to include components from each leadtime within the window, which brings the dimension of the state space up to $\sim 2 \times 10^7$. We also augment the observation vector to include observations from each observation time. Since we are assimilating observations from the first three days of each run, the dimension of the observation space becomes 219.

The specific method whose analysis step we borrow is called the ensemble transform Kalman filter (ETKF), and here the description from Asch, Bocquet, and Nodet, 2016 is outlined. In the ETKF method, the analysis is performed in ensemble subspace rather than in the state space. Algebraically this means expressing the increment as a linear combination of ensemble perturbations

$$\mathbf{X}_f = \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_1^f - \bar{\mathbf{x}}^f, \mathbf{x}_2^f - \bar{\mathbf{x}}^f, \dots, \mathbf{x}_N^f - \bar{\mathbf{x}}^f \right) \quad (2.16)$$

$$\mathbf{x}^a = \bar{\mathbf{x}}^f + \mathbf{X}_f \mathbf{w}^a \quad (2.17)$$

Where $\mathbf{x}_i^f$ is the state of ensemble member i prior to assimilation and the ensemble mean $\bar{\mathbf{x}}^f$ is used as the background state. From Asch, Bocquet, and Nodet (2016) pp.163-164, the optimal coefficient vector $\mathbf{w}^a$ is given by

$$\mathbf{w}^a = \left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-1} \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{d}_b^o \quad (2.18)$$

Where $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\bar{\mathbf{x}}^f$. This method is algebraically equivalent to working in state space with the background error covariance matrix given by the forecast error covariance matrix

$$\mathbf{P}_f = \mathbf{X}_f \mathbf{X}_f^\top \quad (2.19)$$

This matrix is rank deficient since its column space has dimension equal to the ensemble size, but working in ensemble space has considerable advantages over working in state space in terms of computational cost (Hunt, Kostelich, and Szunyogh, 2007). Note that this method only gives a single analysis state, where we started with an ensemble. To get a full analysis ensemble, we first define the analysis error covariance matrix, which is prescribed by Kalman theory (Asch, Bocquet, and Nodet, 2016 section 3.4.3) to be

$$\mathbf{P}^a = (\mathbf{I}_n - \mathbf{K}\mathbf{H}) \mathbf{P}^f \quad (2.20)$$

Where n is the dimension of state space as before. In this case the Kalman gain matrix takes the form $\mathbf{K} = \mathbf{P}^f \mathbf{H}^\top (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1}$. To generate the analysis ensemble, we

need to factorise $\mathbf{P}^a$ to get the analysis ensemble perturbations $\mathbf{P}^a = \mathbf{X}_a \mathbf{X}_a^\top$. We can rewrite equation 2.20 as

$$\begin{aligned}\mathbf{P}^a &= \left(\mathbf{I}_n - \mathbf{X}_f \mathbf{X}_f^\top \mathbf{H}^\top \left(\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R} \right)^{-1} \mathbf{H} \right) \mathbf{X}_f \mathbf{X}_f^\top \\ &= \mathbf{X}_f \left(\mathbf{I}_N - \mathbf{X}_f^\top \mathbf{H}^\top \left(\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R} \right)^{-1} \mathbf{H} \mathbf{X}_f \right) \mathbf{X}_f^\top\end{aligned}$$

So we define $\mathbf{X}_a$ as

$$\mathbf{X}_a = \mathbf{X}_f \left(\mathbf{I}_N - \mathbf{X}_f^\top \mathbf{H}^\top \left(\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R} \right)^{-1} \mathbf{H} \mathbf{X}_f \right)^{\frac{1}{2}} \mathbf{U} \quad (2.21)$$

Since the matrix square root is only defined up to an arbitrary orthogonal matrix, we make this explicit by including an arbitrary orthogonal matrix $\mathbf{U}$ in the form of $\mathbf{X}_a$. This formula can be simplified to

$$\mathbf{X}_a = \mathbf{X}_f \left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-\frac{1}{2}} \mathbf{U}$$

using the Sherman-Morrison-Woodbury identity (Gene H. Golub, 1996 p. 50). The full analysis ensemble can then be generated from the analysis state derived in equation 2.18 via

$$\mathbf{x}_i^a = \mathbf{x}^a + \sqrt{m-1} \mathbf{X}_f \left[\left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-\frac{1}{2}} \mathbf{U} \right]_i \quad (2.22)$$

Where the notation $[_]_i$ refers to the i^{th} column of the matrix being taken. Now the analysis ensemble perturbations match exactly the form of the forecast ensemble perturbations (cite):

$$\mathbf{X}_a = \frac{1}{\sqrt{N-1}} (\mathbf{x}_1^a - \mathbf{x}^a, \mathbf{x}_2^a - \mathbf{x}^a, \dots, \mathbf{x}_N^a - \mathbf{x}^a) \quad (2.23)$$

2.2.3 Practical Aspects

When calculating $\mathbf{P}^f$ from the ensemble, spurious covariances may arise at long distances, especially when the ensemble size is small. This is problematic as it can lead to large increments far from observation locations, which is physically unrealistic. One way to fix this is called localisation, and involves multiplying each component of the background error matrix $\mathbf{P}_{ij}^f$ by a factor which depends on the distance between gridpoints i and j . Representing this distance with $d(i, j)$, we have the following equation:

$$\mathbf{P}_{ij}^{f'} = \mathbf{P}_{ij}^f f(d(i, j)) \quad (2.24)$$

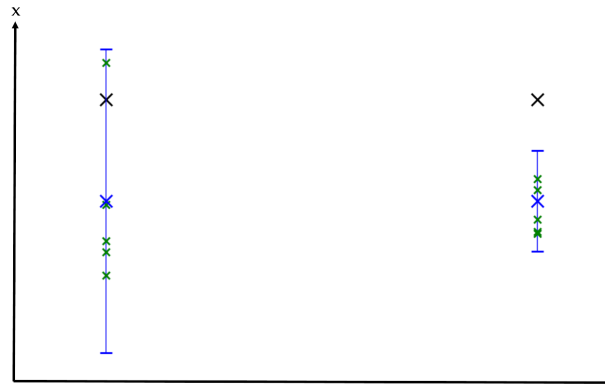


FIGURE 2.3: Left: an ensemble with appropriate spread. Right: an overconfident ensemble. Truth is shown in black, ensemble mean and spread is shown in blue and ensemble members are shown in green.

The function f should be close to 1 when $d(i, j)$ is small and close to 0 when $d(i, j)$ is large, to suppress covariances at long distances while allowing them to exist at short distances (FIND SOURCE)

Another problem that may be encountered is $\mathbf{P}^f$ being an underestimate of the true background error covariance. This is a result of sampling error that affects the estimation of $\mathbf{P}^f$ from the ensemble (cite). This alone is not necessarily undesirable, since it will lead to the analysis covariance $\mathbf{P}^a$ being smaller which reduces spread in the analysis ensemble. Ensemble spread is a commonly used metric of forecast uncertainty, so some reduction in spread is desirable.

It is possible for $\mathbf{P}^f$ to be too small, however, as by equation 2.14 a small $\mathbf{P}^f$ to increments which are erroneously small, and tends to compound over multiple analysis steps since a tighter analysis ensemble leads to a smaller forecast error covariance matrix at the next analysis step, which leads to a problem called filter divergence (Jazwinski (1970), pp.301-305). Note that in our case it will not compound since we are only doing one analysis step per model run, but it should be checked that the increments are realistic and the analysis ensemble spread is not too small even after one analysis step.

We may test this by comparing the analysis ensemble to the true value. If the truth lies outside the range given by the ensemble spread, then the ensemble spread is too small and we say the ensemble is ‘overconfident’. A schematic is shown in figure 2.3. The ensemble on the left has appropriate spread, as the ensemble mean lies within the range given by the ensemble. The ensemble on the right is overconfident, as the range of the ensemble is too small to capture the truth.

An overconfident ensemble is undesirable, since as mentioned before, ensemble spread is commonly used to estimate uncertainty in a forecast. An overconfident ensemble may lead to too much confidence being placed in an incorrect forecast, which is detrimental to any decision making that is informed by the forecast.

Various methods are available to counteract this problem, but one of the simplest is called inflation (Anderson and Anderson, 1999). This involves multiplying the forecast error covariance matrix $\mathbf{P}^f$ by a predefined factor $\alpha > 1$ prior to the analysis step. This increases the size of the increments that may be applied by the system, and also increases the spread of the analysis ensemble, which reduces overconfidence.

Chapter 3

Experimental Setup

This chapter is intended to describe the details of the experimental setup used in this project. It contains a description of the wave model used to generate the forecasts, as well as an overview of the background state, the observations used and the errors present in these observations. Finally a description is given of the twin experiment method used in this project, as well as the data assimilation setup.

3.1 Model Description and Domain

Waves are modelled by Fugro using WW3 (WW3DG, 2019), which uses as its base spectrum a wavenumber-direction spectrum $F(k, \theta)$, rather than the frequency-direction spectrum described in section 2.1.3. The spectrum is discretised using 29 sampled wavenumbers k and 24 angles θ . The wavenumber-direction spectrum is related to the frequency-direction spectrum by the Jacobian transform

$$F(\sigma, \theta) = \frac{\partial k}{\partial \sigma} F(k, \theta) = \frac{1}{c_g} F(k, \theta)$$

Wave propagation is then described by the equation

$$\frac{D}{Dt} N(k, \theta) = \frac{S}{\sigma} \quad (3.1)$$

Where $N(k, \theta) = F(k, \theta)/\sigma$ as before and S describes the net effect of sources and sinks for the spectrum $F(k, \theta)$.

In deep water, S consists of three main parts: an atmospheric interaction term S_{in} , a nonlinear wave-wave interaction term S_{nl} and a dissipation term S_{ds} .

S_{in} is dominated by an exponential wind-wave growth term, so S_{in} is equal to some coefficient multiplied by $\sigma N(k, \theta)$. This leads to a feedback which is usually positive but can be negative in the case of swell, and represents the main source of wave energy.

S_{ds} is generally dominated by wave breaking, and it represents a sink of wave energy. Similar to S_{in} , it is equal to a negative coefficient multiplied by $\sigma N(k, \theta)$. The balance between S_{in} and S_{ds} determines the overall growth or decay of wave energy.

S_{nl} represents interactions between waves due to nonlinear processes, which results in energy transfer between waves of different frequencies. (maybe expand - Janssen, 2004)

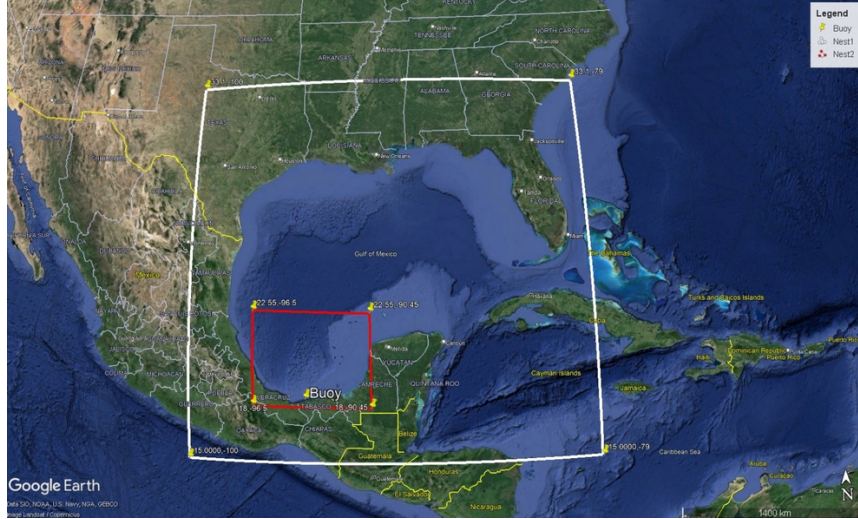


FIGURE 3.1: Model domains shown on map.

In shallow water, processes such as wave-bottom interactions, wave breaking and reflection off shorelines become important, so additional source terms are included to describe this.

From the forecasted spectra, other variables such as wave height, period and mean direction are derived. The three variables we will be focusing on in this study are significant wave height H_s , and two measures of the dominant wave period T_{m01} and T_{m02} . These are related to the spectrum via the spectral moments (equations 2.8, 2.11).

The reason for assimilating both T_{m01} and T_{m02} is that they are defined from different spectral moments, so assimilating both will provide more information about the spectrum than either one alone, which we hope will lead to better DA results.

As forcing, WW3 uses just the 10 metre wind, and this comes from the National Oceanic and Atmospheric Association’s (NOAA) global ensemble forecasting system (GEFS). A 31 member ensemble is taken from GEFS, and the u and v wind fields at 10m are used to force WW3. Each GEFS ensemble member provides forcing for one of the 31 WW3 ensemble members

The domain in which forecasts are produced lies between $15.0 - 33.0^\circ\text{N}$ and $79.1 - 100^\circ\text{W}$ in the Gulf of Mexico. Fugro use a nested domain to produce the forecasts, running WW3 on a 0.5° resolution global rectangular grid, and two higher resolution nested grids covering the region of interest. There is a 0.1° resolution grid spanning the region described above, and a 0.05° grid spanning from $18 - 22.5^\circ\text{N}$ and $90.5 - 96.5^\circ\text{W}$. These domains are shown on a map in figure 3.1.

Forecasts are produced at 12Z each day, and run up to seven days ahead. Seven of these daily runs are available with start dates spanning from 06/03/2022 to 12/03/2022. A schematic of this process is shown in figure 3.2. As stated in section 2.2.2, we are unable to run the forecast model in this project, so each 7-day forecast window is treated separately with a single analysis step.

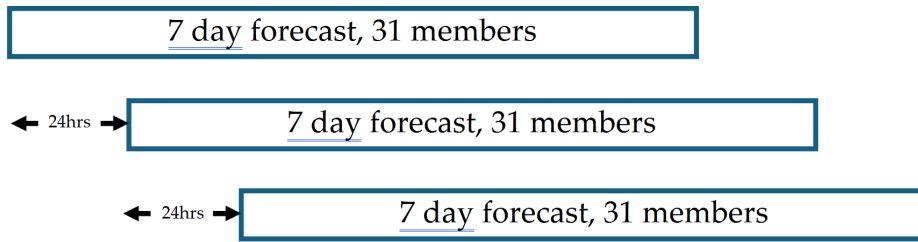


FIGURE 3.2: Schematic of the forecasting setup used by Fugro. A new 7-day 31-member ensemble forecast is produced at 12Z each day.

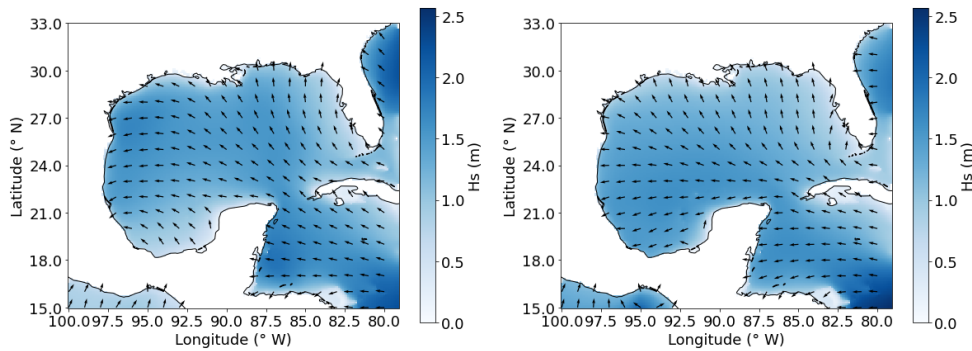


FIGURE 3.3: Left: H_s (colours) and wave direction (arrows) averaged between 12Z 06/03/2022 and 12Z 12/03/2022. Right: The same averaged between 12Z 13/03/2022 and 12Z 19/03/2022

3.2 Background State

In this section we present a general overview of the background state in the domain, coming from the forecast data which spans a period from 06/03/2022 to 19/03/2022. We identify the broad features of the waves in this window, and also discuss the effect of a cold front that passed through the region on 12/03/2022.

Figure 3.3 shows the significant wave height and direction averaged over the first six days and last six days of the forecast period. H_s is mostly between 1m and 2m in the central gulf, with lower wave heights of < 1 m in sheltered coastal areas such as the western side of Florida and the Yucatan peninsula. The central part of the domain has largely east-southeasterly waves over both periods, which diverge after they pass between Yucatan and Cuba. The two periods are separated by the passage of strong cold front (figure 3.4) and there is a slight shift to more easterly waves after the front has passed. This change is likely related to the wind backing associated with the front. The southern part of the Gulf of Mexico has slightly higher waves, around 1 - 1.5m, in the six day period after the passage of the front.

There is a large peak in significant wave height on the 12/03/2022, related to a cold front passing through the region. A map of H_s in the GoM at 18:00 on 06/03/2022 is shown in figure 3.4, along with a surface chart from the same time. The cold front is visible on the chart, and high waves are present in a large region behind the front. This is in line with Appendini et al. (2014), a study into the wave climatology of the GoM. In that paper, cold fronts are identified as a common driver of high waves between October and April, with the most intense peaks occurring between December and March.

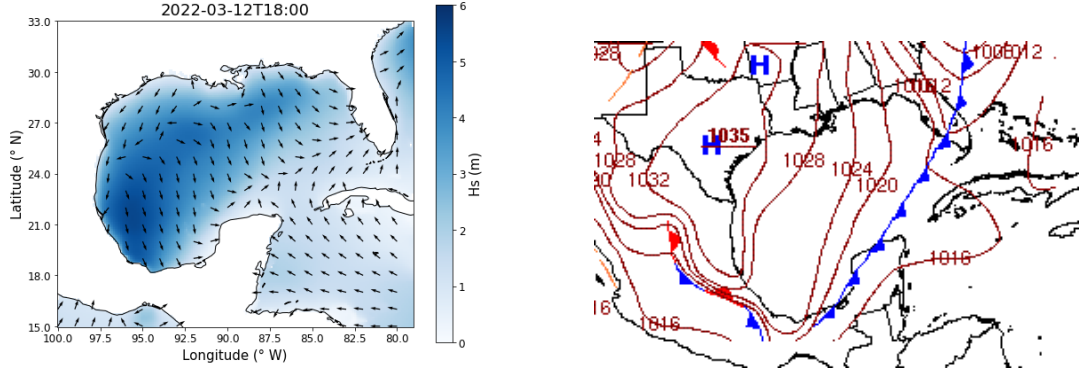


FIGURE 3.4: Left: H_s (colours) and direction (arrows) in the GoM at 18Z on 12/03/2022. Right: Surface analysis chart valid for 18Z on 12/03/2022, taken from NOAA WPC surface analysis archive

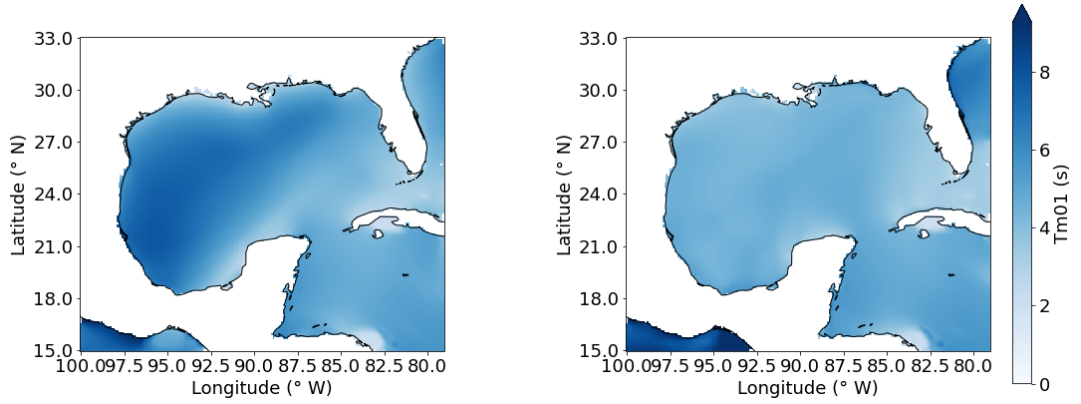


FIGURE 3.5: Left: T_{m01} at 18Z on 12/03/2022. Right: T_{m01} averaged over the period from 12Z 10/03/2022 to 12Z 12/03/2022

Figure 3.5 shows T_{m01} at 18Z 12/03/2022 and averaged over the preceding 48 hours. The high waves associated with the cold front at 18Z have a period of ~ 8 s, which is higher than the period in the rest of the GoM at that time and also higher than the waves across the entire GoM in the 48 hours prior.

3.3 ADCP Observations

Our observations come from an acoustic doppler current profiler (ACDP) buoy, which estimates the spectrum by measuring the wave induced velocity and surface height variations. According to Vallis, 2017, the velocity and surface height anomaly induced by a simple wave with wave vector $\mathbf{k} = (k_x, k_y)$, frequency σ and amplitude a is

$$\begin{aligned} u &= a \frac{k_x}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ v &= a \frac{k_y}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ w &= -ia \frac{k}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \sinh k(z + D) \\ \eta &= a \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \end{aligned}$$

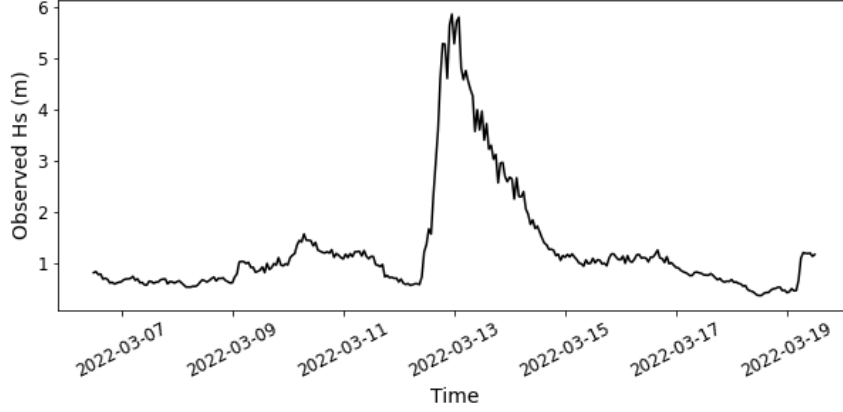


FIGURE 3.6: Observed timeseries of significant wave height from ADCP buoy

where only the real part of each expression should be taken. Note that the horizontal velocity (u, v) is always in the same direction as the wave vector $\mathbf{k}$. From the ADCP measurements, a timeseries of wave induced energy per unit area $E = \int_{-D}^{\eta} \frac{1}{2} \rho_0 \|\mathbf{u}\|^2 dz + \frac{1}{2} \rho_0 \eta^2$ can be calculated. This is equivalent/a good approximation to the wave energy in equation 2.6 (source) so a Fourier transform of E gives the wave spectrum integrated across all directions $E(\sigma) = \int F(\sigma, \theta) d\theta$ (how is directional distribution estimated?)

Other variables are then deduced from this spectrum, including H_s , T_{m01} and T_{m02} as described above.

We have observations available over a roughly 6 month period spanning from October 2021 to April 2022, which includes the period for which forecasts are available. A timeseries of the observed significant wave height from the forecast period is included in figure 3.6. The peak in wave height late on the 12th of March is clearly visible on the plot.

Figure 3.7 shows a timeseries of observed H_s and T_{m01} from a longer period. We see that outside of specific peaks, H_s is generally $< 1\text{m}$. This is consistent with the model background state which showed H_s in the buoy region to be $< 1\text{m}$ for much of the forecast period. The peaks generally consist of a sharp increase in H_s followed by a more gradual decrease down to the baseline level. This is consistent with the slightly higher waves seen in the model background state in the buoy region after the passage of the front. As seen in the background state, the peaks in H_s are associated with peaks in T_{m01} . This points to a positive correlation between H_s and T_{m01} which is discussed further in section 4.1.

Figure 3.8 shows a density plot of observed H_s and wave direction at the buoy location. There is a clear peak in frequency corresponding to waves of height $\sim 0.5\text{m}$ coming from a northeasterly direction. Some higher waves (up to $\sim 5\text{m}$) are present coming from a north-northwesterly direction. This lines up with the direction of waves produced by the cold front on 12/03/2022, suggesting that the waves produced by this kind of event consistently reach the buoy from a north-northwesterly direction. Since the buoy is near the coast in the southern part of the gulf, it is sheltered from the south so there are virtually no waves coming from this direction.

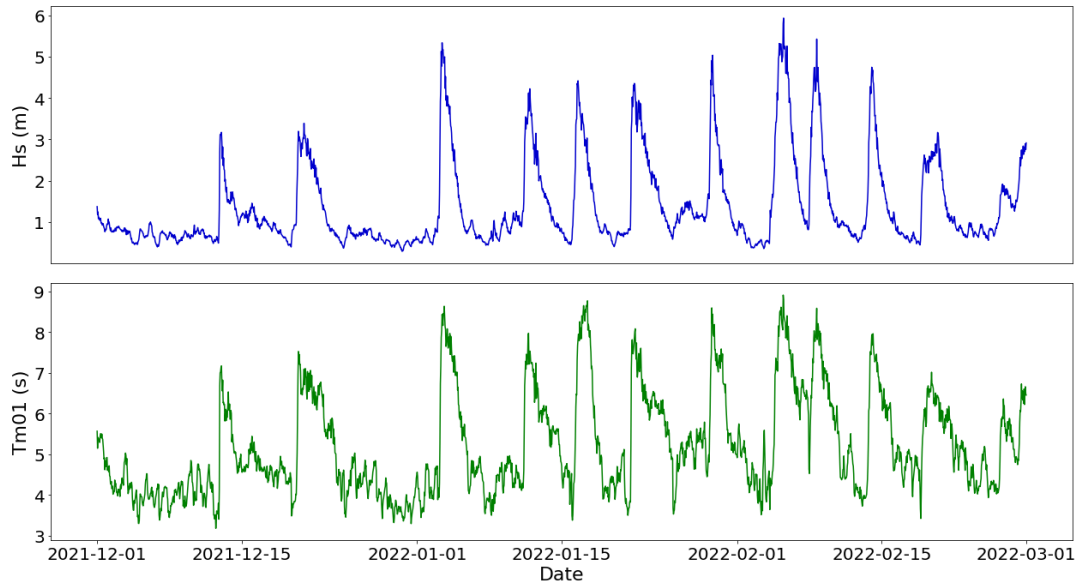


FIGURE 3.7: Timeseries of observed H_s (top) and T_{m01} (bottom) over a three-month period from 01/12/2021 to 01/03/2022

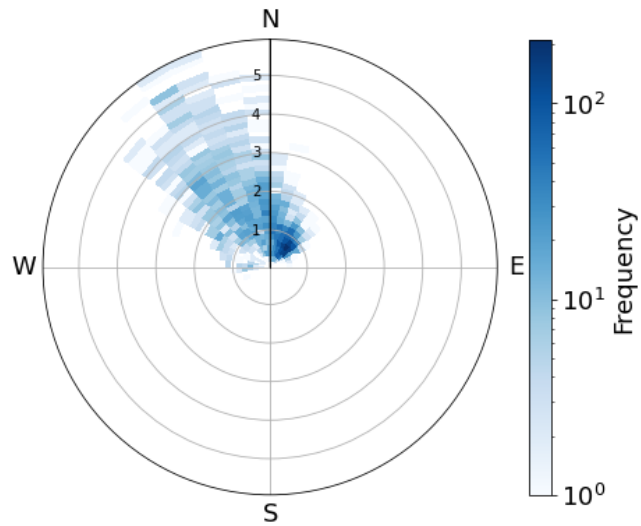


FIGURE 3.8: Density plot of H_s (radial axis) and wave direction at the buoy location.

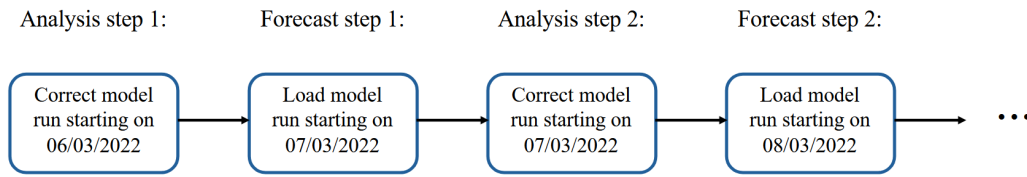


FIGURE 3.9: Schematic of the forecast and analysis steps from the ETKF scheme employed in this project.

3.4 Twin Experiment

In this study, we will not be using real observations in our assimilation but setting aside one ensemble member to represent the ‘truth’ and generating fictional observations based on this forecast data. This is known as a twin experiment, and has several advantages over assimilating real observations in this context:

- The ‘truth’ is available away from the location of observations, so forecast error can be calculated over the whole model domain. This allows the effectiveness of DA to improve forecasts away from observation locations to be evaluated

The observations are generated by sampling values from the truth at predefined observation locations and times. Here we sample hourly values from the nearest gridpoint to the buoy to match the real observations that are available. To represent observation error, some random noise is added to the sampled values. This noise is drawn from a normal distribution with standard deviation equal to the error standard deviation derived in section 3.3

A downside of using the twin experiment method in this project is its inability to correct biases in the model. Figure 4.7 shows a possible bias in the forecasted values of H_s , which would be desirable to try to correct with DA. Since the data that we assimilate in this project is also from model output, it will be subject to all the same biases as the forecast data so it is not possible for the DA to correct these biases.

3.5 Data Assimilation Setup

To perform the data assimilation we use pyPDAF (Chen, Nerger, and Lawless, 2024), which is a python interface for the fortran-based Parallel Data Assimilation Framework (PDAF). We will use the ETKF method which comes builtin with PDAF. As we are not able to run the WAVEWATCH III model ourselves, at each analysis step we will correct one 7-day forecast window all at once, and then the forecast step will consist of selecting data from the next 7-day window. In this way the time within the 7-day window is treated like another spatial coordinate by the DA system. A schematic of this process is included in figure 3.9

At each analysis step, we will assimilate observations from the first three days of the forecast window. This allows us to investigate errors at times after the assimilation window and assess the capability of DA to improve the forecast at future time when observations are unavailable.

Chapter 4

Background and Observation Errors

This chapter contains a discussion of the background and observation errors in our DA system. Though the background are determined directly from the ensemble when using the ETKF method, understanding the form they take is crucial to understanding how the DA system works. This is because, according to equation 2.15, the DA increments always lie in the column space of the background error covariance matrix. This means that the background errors determine how information from the observations can be propagated around the domain by DA. The observation errors are also important to understanding a DA system since the balance between observation and background errors determines how much weight is given to the observations by the DA. The observation errors must be estimated manually, and a description of the methods used to do this are included here.

4.1 Background Errors

The background errors are a crucial part of any DA system since they determine how information from the observations can be spread around the domain. From equation 2.15, the increments always lie in the column space of the background error covariance matrix $\mathbf{B}$. This means that understanding the form $\mathbf{B}$ takes is vital to understanding the performance of the DA system. In this section, we investigate the background error by examining the ensemble variance and forecast error covariances.

The ensemble variance is a good measure of the general scale of the background error, since in our DA method the diagonal elements of the forecast error covariance matrix are simply the ensemble variances at each gridpoint and timestep.

Figure 4.1 shows a timeseries of H_s ensemble mean and variance at the buoy location from the forecast starting 12Z 10/03/2022. We see the peak on the 12th as seen in figure 3.4. We also see that the variance is a strong function of the mean. The variance in H_s peaks just before the peak in the mean, and drops back to a lower value as the mean reaches its peak. This is notable since the diagonal elements of our forecast error covariance matrix $\mathbf{P}^f$ are simply the ensemble variances at each gridpoint and timestep, so ensemble variance is a good measure of the magnitude of the background error. This means that the background error in H_s is highest just before the peak in H_s .

Figure 4.2 shows the same as 4.1 but for T_{m01} . Again we see a peak in the variance just before the peak in the mean. In this case peak in T_{m01} is preceded by a dip, and

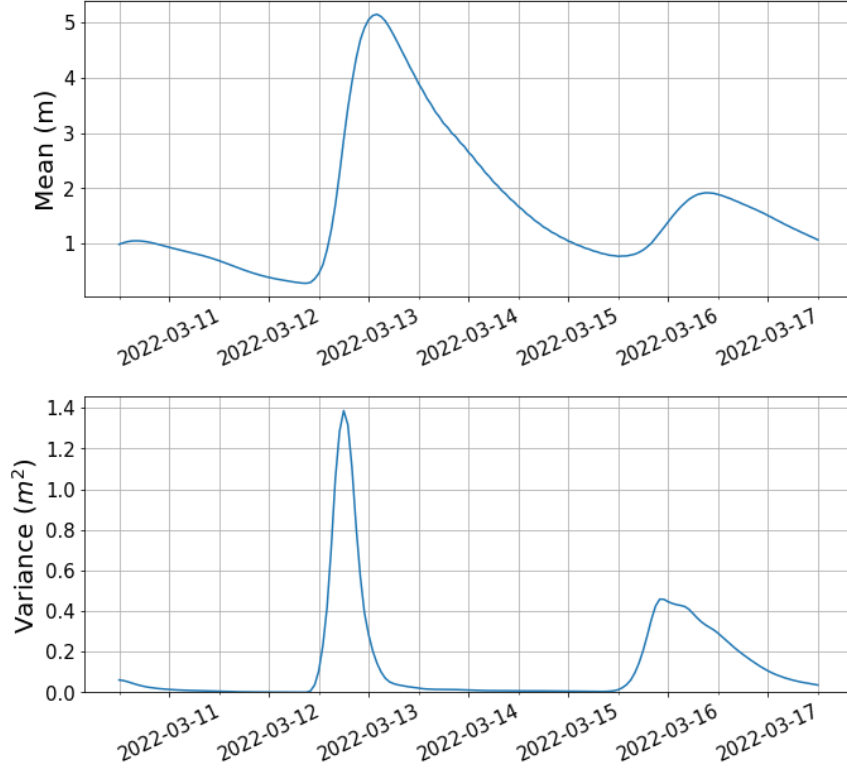


FIGURE 4.1: Timeseries of forecast H_s ensemble mean and variance at the buoy location from the 10/03/2022 model run.

this dip is also associated with a peak in the ensemble variance. This means that the background error in T_{m01} is high for a period of ~ 24 hours, as opposed to ~ 12 hours for H_s .

In the ETKF method, the background error covariance matrix is approximated by the forecast error covariance matrix $\mathbf{P}^f$. The elements of $\mathbf{P}^f$ are the ensemble covariances at each pair of gridpoints:

$$\mathbf{P}_{ij}^f = \text{cov} \left(\mathbf{x}_i^f, \mathbf{x}_j^f \right) \quad (4.1)$$

Where $\mathbf{x}_i^f$ refers here to the ensemble values in component i of the state vector. In our case this corresponds to the values from some specified gridpoint and timestep. Here we present maps of these covariances in order to illustrate the forms these covariances can take and gain insight into the workings of our DA system. Arrows showing wind/wave direction will be used to explain the form the covariances take.

Figure 4.3 shows the spatial covariance of H_s at 12Z 12/03/2022. The colours on the map show the covariance between H_s at each point in the domain and H_s at the buoy (marked with a cross). The arrows show the direction and magnitude of the wind, averaged over the preceding 24 hours. We see that the strongest covariances are confined to the western GoM and do not extend very far from the buoy. From the wind arrows we see that the wind in the region of strong covariance is very low, similar to the wind at the buoy, and that the wind in regions of weak covariance is high.

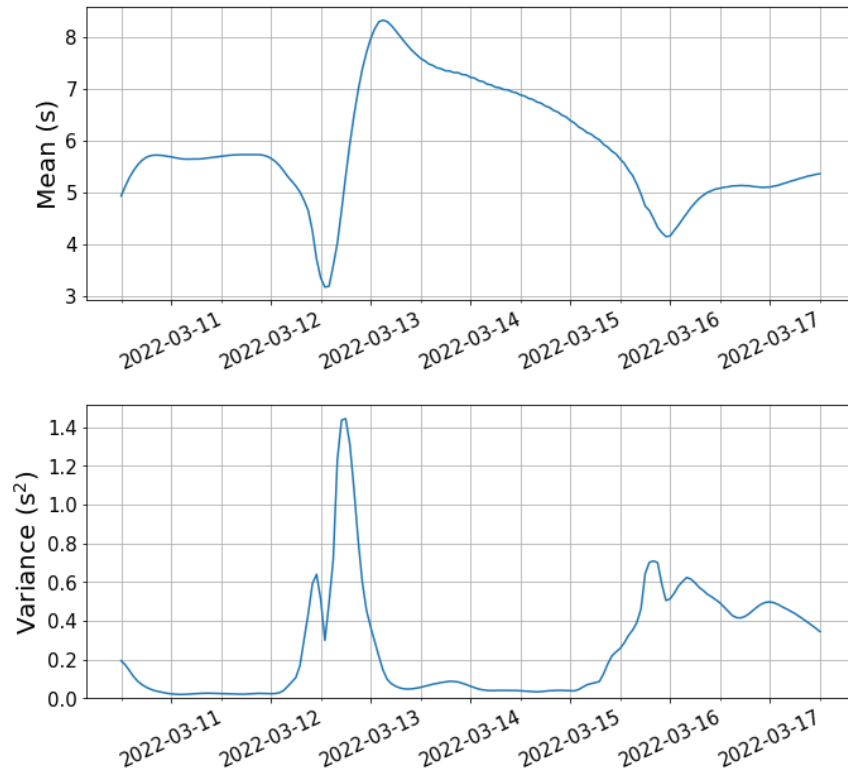


FIGURE 4.2: Timeseries of forecast T_{m01} ensemble mean and variance at the buoy location from the 10/03/2022 model run.

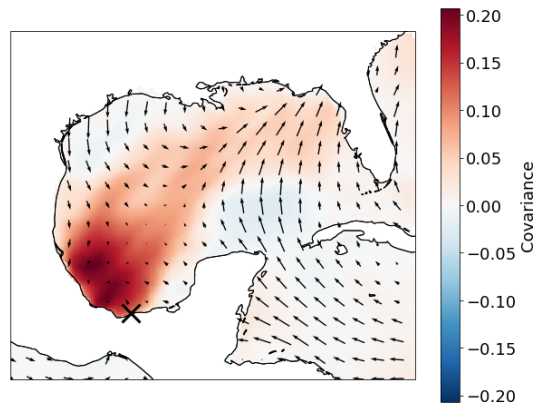


FIGURE 4.3: Colours showing spatial covariance of H_s at 12Z 12/03/2022. Arrows showing wind averaged over prior 24 hours.

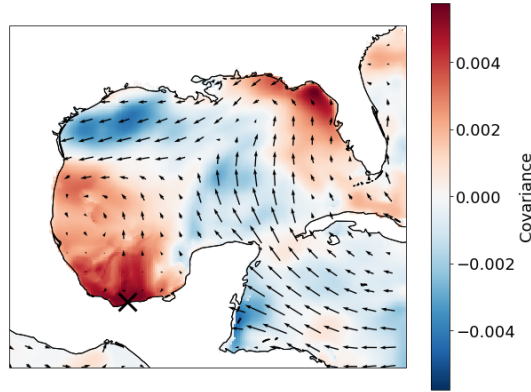


FIGURE 4.4: Colours showing spatial covariance of H_s at 12Z 11/03/2022. Arrows showing wind averaged over prior 24 hours.

This suggests that the wind forcing plays a role in determining the nature of the ensemble covariances. This is unsurprising since according to Lefèvre and Aouf, 2012, ocean wave prediction is primarily a forcing problem, and so we expect the nature of the forcing to play a major role in determining the properties of the waves. This is our motivation for averaging the wind values over the preceding 24 hour window, since the wind forcing takes time to have an impact on the waves.

The wind forcing can also induce covariances at longer distances. Figure 4.4 shows the same as 4.3 but at 12Z 11/03/2022. We see positive covariance near the buoy as before, but there is also a patch of positive covariance in the northeastern GoM, which is a long way from the buoy. This is likely because the wind in this region is weak, similar to the wind in the vicinity of the buoy. Here we also see some areas of negative covariance in regions of strong winds.

It is also possible that these long distance covariances are physically unrealistic. As discussed in section 2.2.3, an inherent limitation in estimating the background error from the ensemble is that a finite ensemble size leads to sampling error in $\mathbf{P}^f$. This can cause spurious covariances to arise at long distances (Hunt, Kostelich, and Szunyogh, 2007), which can be fixed with localisation. This possibility is not explored in this study, partly due to the fact that some of the long-distance covariances appear to be genuine, but this is something that should be included in future work on the topic.

The scale of covariance seen here is much smaller than that seen in figure 4.3. From figure 4.1 we see that the variance in H_s at 12Z 11/03/2022 is much smaller than the variance at 12Z 12/03/2022, which likely explains the differing scales in covariance.

Another thing that can cause long-distance covariances is travelling swell waves. Figure 4.5 shows the H_s covariance at 18Z 13/03/2022 with a one day lag. This means the H_s ensemble at the buoy is sampled at 18Z 13/03/2022 and the H_s ensemble elsewhere in the domain is sampled at 18Z 12/03/2022, to show the correlation with earlier sea states. We see a large region of positive covariance spanning the Western GoM and much of the Northern part also. The arrows show the wave direction at 18Z 13/03/2022, and this tells us that waves from the Northwestern GoM travel West and South towards the buoy. This means they are able to influence the sea state at the buoy, which explains the positive covariance.

So far we have only examined the covariance of H_s with itself. There are also covariances present between the different variables assimilated in this study, and this is

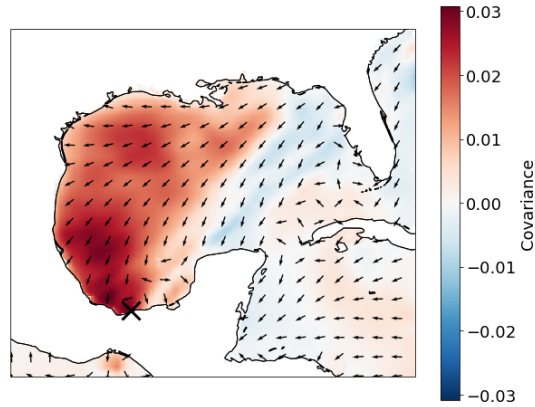


FIGURE 4.5: Colours showing covariance of H_s at buoy at 18Z 13/03/2022 with H_s elsewhere at 18Z 12/03/2022. Arrows showing wave direction at 18Z 12/03/2022

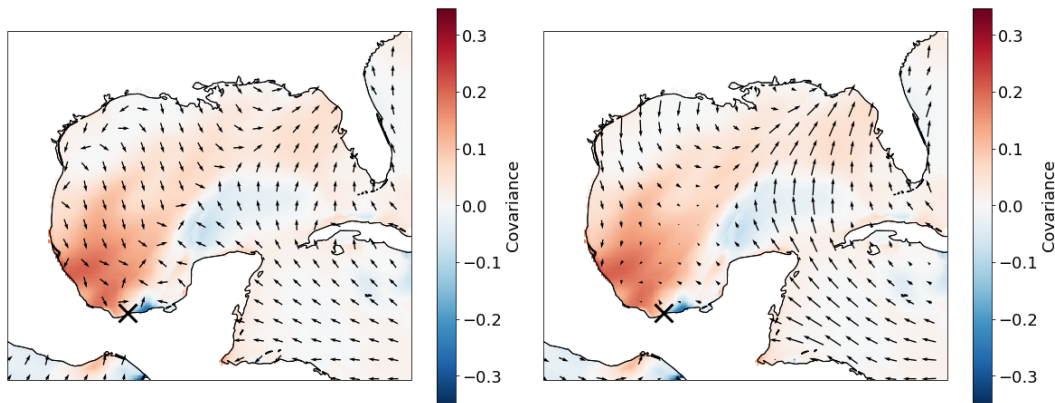


FIGURE 4.6: Cross-covariance between H_s at the buoy location and T_{m01} elsewhere at 12Z 12/03/2022. Arrows show wave direction (left) and wind (right)

important for the DA. These cross-covariances allow assimilation of one variable to affect other variables than the one assimilated. This will allow the assimilation of H_s alone to impact T_{m01} and T_{m02} , and allow the assimilation of all three variables to produce different forecasts of H_s than would be obtained by assimilating H_s alone. It is hoped that these cross-covariances will improve DA performance by allowing each observation assimilated to provide more information than it would in their absence.

Figure 4.6 shows the cross-covariance between H_s at the buoy and T_{m01} elsewhere at 12Z 12/03/2022. We see that there is a large area of positive covariance to the Northwest of the buoy, with the wave direction (left plot) pointing mostly towards the buoy in this region. This suggests that the positive covariance may be related to travelling swell waves, since large swell waves with long period in the region Northwest of the buoy could travel towards the buoy and lead to higher waves there. However more investigation is required to determine if this is actually the case.

There are also small patches of strong negative covariance, directly East of the buoy and also South in the Pacific ocean. Since there is no obvious difference in the winds in each region to explain this, these negative covariances could be spurious. This is especially true of the negative covariance seen in the Pacific, since there is no way for the waves here to interact directly with the waves at the buoy.

Note also that the scale of positive covariance seen here (approximately 0.15) is smaller than the scale of covariance seen between H_s and itself at the same time (figure 4.3, approximately 0.2). This means that observations of H_s will have less impact on T_{m01} than observations of T_{m01} will and vice versa, which is what we intuitively expect.

In this section, we have presented maps showing the forms that the forecast error covariances can take, and attempted to explain them by referring to physical explanations from winds and wave directions, and also to sampling error. However, the physical explanations discussed here are quite simple and surface-level, and more investigation would be useful to explain the form of the covariance. Since understanding the physical mechanisms behind the covariances is not completely central to this project's goal of explaining DA performance, it is not explored further here, but it would be a good avenue for future work.

4.2 Diagnosis of Observation Errors

To simplify the computation, we assume that the observation error covariance matrix $\mathbf{R}$ is diagonal, essentially ignoring covariances between observations taken at different times. This also makes determining $\mathbf{R}$ easier since all that is required is to estimate the observation error variance at each point. To determine the magnitude of the observation errors, several methods are available. A simple one is to plot a histogram of observation - ensemble mean, called an $o - b$ plot. The spread in $o - b$ comes from a combination of observation and background error, so a simple way of estimating the observation error is to assume that observation error accounts for a constant proportion of this error, say 50%. An $o - b$ histogram for significant wave height is shown in figure 4.7. From the plot we see that the spread on the histogram is approximately 1m, so the standard deviation of observational errors in significant wave height may be estimated to be approximately 50cm.

Another thing that is visible from the $o - b$ plot is a bias in either the H_s forecasts or observations. The peak of the histogram is centred at around 30cm, which means that the forecast model tends to predict wave heights that are about 30cm less than those observed. If this bias is a result of the model, it would be desirable to investigate the capability of DA to reduce this bias. However, with the twin experiment method that is used in this project, this is not possible (see section 3.4 for details).

The histogram approach may be quantified more precisely following Desroziers et al., 2006, by defining $d_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}_b$ and using the relation

$$\mathbb{E} \left[d_b^o (d_b^o)^\top \right] = \mathbf{H}\mathbf{P}^f\mathbf{H}^\top + \mathbf{R} \quad (4.2)$$

Taking traces and dividing by the number of observations, we get a good estimate for the observation error variance which is

$$\frac{\mathbb{E} \left[\sum_i (d_b^o)_i^2 \right]}{\# \text{ observations}} - \frac{\text{Tr} (\mathbf{H}\mathbf{P}^f\mathbf{H}^\top)}{\# \text{ observations}} \quad (4.3)$$

In our case, the observation operator is very simple and simply selects the forecast values from the appropriate location and times, so the diagonal elements of $\mathbf{H}\mathbf{P}^f\mathbf{H}^\top$ are simply the ensemble forecast variances at the observation location and times.

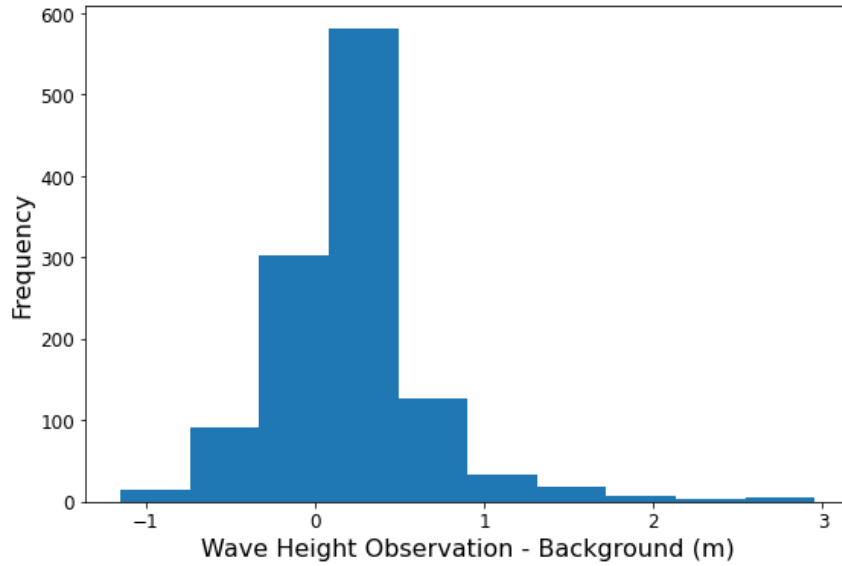


FIGURE 4.7: Observation - background histogram for significant wave height.

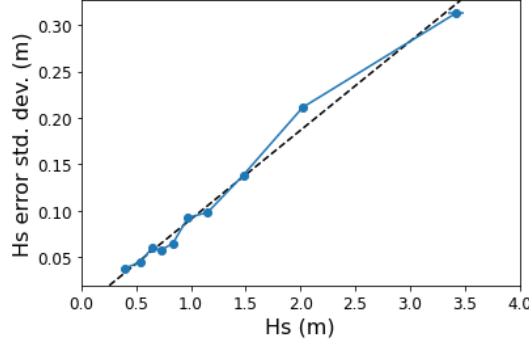
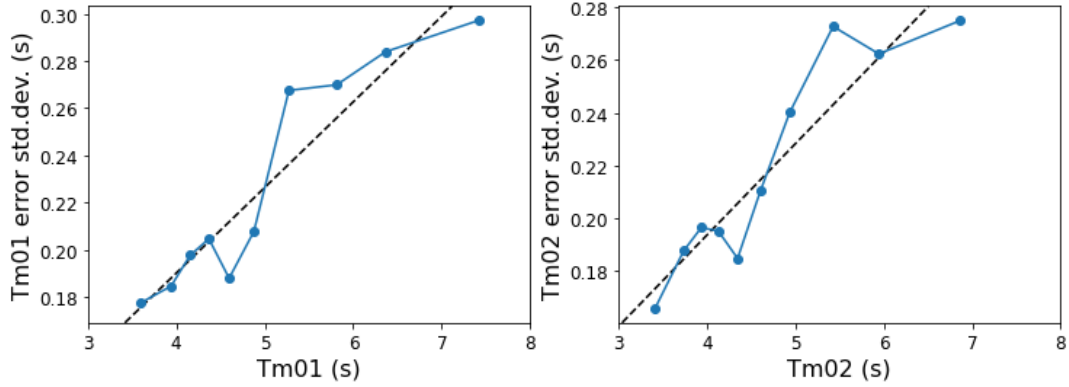
Another available method is that used in Houghton et al., 2023. They used a collocation study to estimate the error in wave height observations from a distributed network of drifting sofar spotter buoys. Pairs of simultaneous observations from locations within 50km of each other were collected and binned according to the average wave height of each pair. The difference of each pair was taken, and the standard deviation of differences from each bin was computed. They found a strong dependence of error on wave height.

In our case, we only have observations from one buoy and they are available hourly, so we take the pairs to be observations one hour apart. We then calculate the standard deviation using the method described above to obtain the plot in figure 4.8. As in Houghton et al., 2023, we find a strong linear relationship between errors and wave height, with the error standard deviation being approximately 9.6% of the wave height.

Recall from figure 4.1 that the ensemble variance H_s peaks just before the peak in H_s and drops back to lower values as H_s is at its highest value. Since ensemble variance is a good measure of the magnitude of the background error, the background error is large compared to observational error just before the peak. This means that observations from this period will be given higher weighting than most. During and shortly after the peak, the opposite is true: background error is small compared to observational error, so observations from this period will be given lower weighting.

Repeating this process with the observations of T_{m01} and T_{m02} yields the plots shown in figure 4.9. The error distributions for T_{m01} and T_{m02} also fall approximately on straight lines, though the fit is not as good as for H_s . The gradients are also shallower, with the error standard deviations being approximately 3.5% of the observed variable for both. So we will model the error standard deviation in H_s as being equal to 10% of H_s , and the error standard deviation in T_{m01} and T_{m02} as 3.5% of T_{m01} and T_{m02} respectively.

Since the errors in period are such a small percentage of the observations, it is possible that at low periods the error may become very small, which would lead to DA

FIGURE 4.8: A plot of H_s error standard deviation against H_s .FIGURE 4.9: Plots of T_{m01} and T_{m02} error standard deviation against T_{m01} and T_{m02} respectively.

system fitting the observations too closely and producing physically unrealistic increments. To avoid this, we set a minimum error threshold ε_0 and set the observation error standard deviation equal to this threshold whenever the scaling method gives a value less than ε_0 . This threshold is determined using the method from Desroziers et al., 2006, using only the observations of wave height with a value of less than 5 seconds, as these are the observations which may lead to small errors.

The terms in 4.3 were calculated to be

$$\frac{\mathbb{E} \left[\sum_i (d_b^o)_i^2 \right]}{\# \text{ observations}} = 0.29s^2, \frac{\text{Tr}(\mathbf{H}\mathbf{P}^f\mathbf{H}^\top)}{\# \text{ observations}} = 0.17s^2$$

So the observation error variance is estimated to be $0.12s^2$ and the standard deviation threshold is set at $0.35s$. Examining the $o-b$ histograms for T_{m01} and T_{m02} in figure 4.10 confirms that this is a sensible value to choose.

In this section, we have found that the observations error standard deviations in all variables are strong functions of the observed values, with the error in H_s being approximately 9.6% of the observed value and the error in both periods being approximately 3.5% of the observation. Motivated by this, we will set the H_s error to be 10% of the observed value in the assimilation of wave height alone. For the assimilation of all variables, there is more risk of fitting the observations too closely as we are assimilating more observations. Because of this, we increase the scales of error

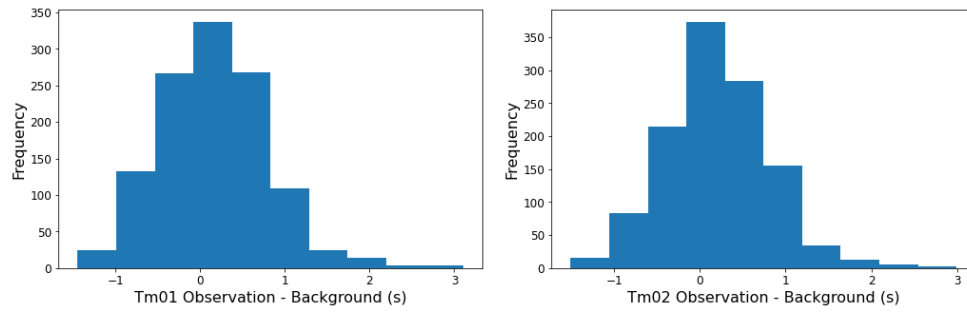


FIGURE 4.10: Observation minus background histograms for T_{m01} (left) and T_{m02} (right).

standard deviation from those found above to 12% for wave height and 7% for the periods. We also implement a minimum error threshold of $0.35s$ for the periods.

Chapter 5

Results

This chapter contains an analysis of the errors in the DA output as compared to the model output with no DA. We also examine the results of assimilating H_s alone (H_s DA) compared to assimilating H_s , T_{m01} and T_{m02} (all vars DA).

For the purposes of a simple visual examination, we include timeseries of ensemble mean from each run, plotted alongside the truth. Figure 5.1 shows this for variables H_s and T_{m01} at the buoy location from the 12Z 09/03/2022 model run. We see little change in H_s during the assimilation window as the no DA forecast matches the truth very closely here. After the end of the window, we see that H_s DA matches truth very well, while all vars DA is close to no DA.

From the T_{m01} plot, we see that H_s is similar to no DA during the window while all vars DA is close to the truth. This is unsurprising as T_{m01} is assimilated directly in all vars DA while H_s is not. This means that any change to T_{m01} from H_s DA relies on cross covariance between the two variables, which is generally small. This indicates that the different variables being assimilated are having the effects we expect them to have. After the window, the performance is more variable: H_s DA does well at 3-4 days, all runs match the truth closely between 4 and days, and none are very close after 5 days. The divergence at 3-4 days and after 5 days is to be expected as this is when the largest errors are present in the original forecasts: 3-4 days because of the large swell event and 5+ days because of the chaos inherent in wave forecasts.

For comparison, figure 5.2 shows the same timeseries at 30° N 80° W, which is outside the GoM in the northern Atlantic ocean. As this location is so far from the buoy, we do not expect to see any kind of improvement from the DA, and indeed this is what we see. All vars DA, H_s DA and no DA are all a similar distance from the truth which indicates that DA has not improved the forecast. One thing to note is that there is often a large difference between the no DA line and both the H_s DA and all vars DA lines, meaning that the increments $\mathbf{d}_b^a = \mathbf{x}_a - \mathbf{x}_b$ are quite large. This is physically unrealistic as we would not expect the ocean state at the buoy location to have any effect on the ocean state at 30° N 80° W. This is explored further in section 5.1.

The divergence of the different runs after the swell event tells us that the DA system has captured the dynamical chaos of the physical wave system.

5.1 Increments

The increments $\mathbf{d}_b^a = \mathbf{x}^a - \mathbf{x}_b$ are the corrections applied to the background state by the DA system. From equation 2.14, the state that minimises the cost function will be close to the background when $\mathbf{P}^f$ is small compared to $\mathbf{R}$. This means that we

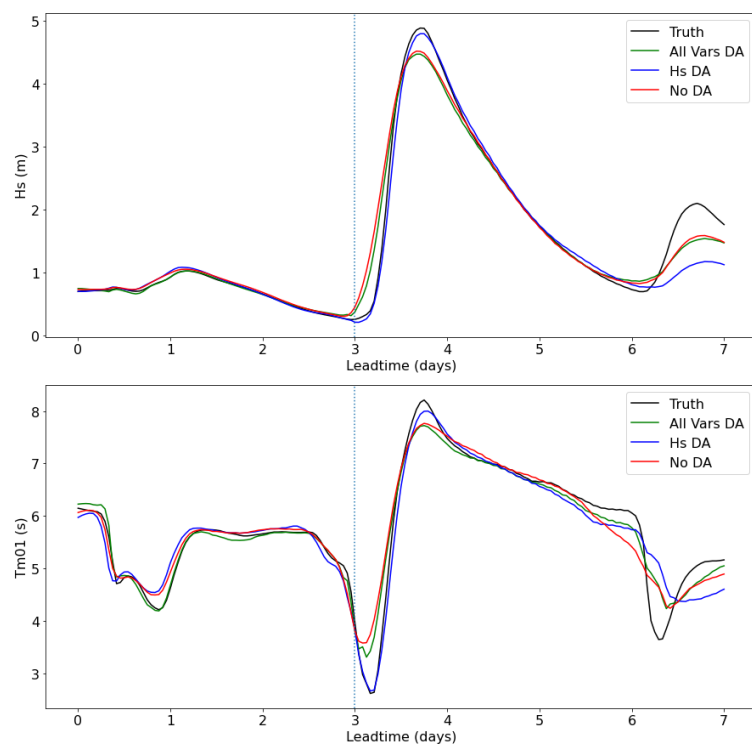


FIGURE 5.1: Timeseries of H_s (top) and T_{m01} (bottom) at buoy location from 12Z 09/03/2022 model run. No DA ensemble mean is shown in red, H_s DA ensemble is shown in blue, all vars DA ensemble mean is shown in green and truth is shown in black. Blue dotted line indicates the end of the assimilation window.

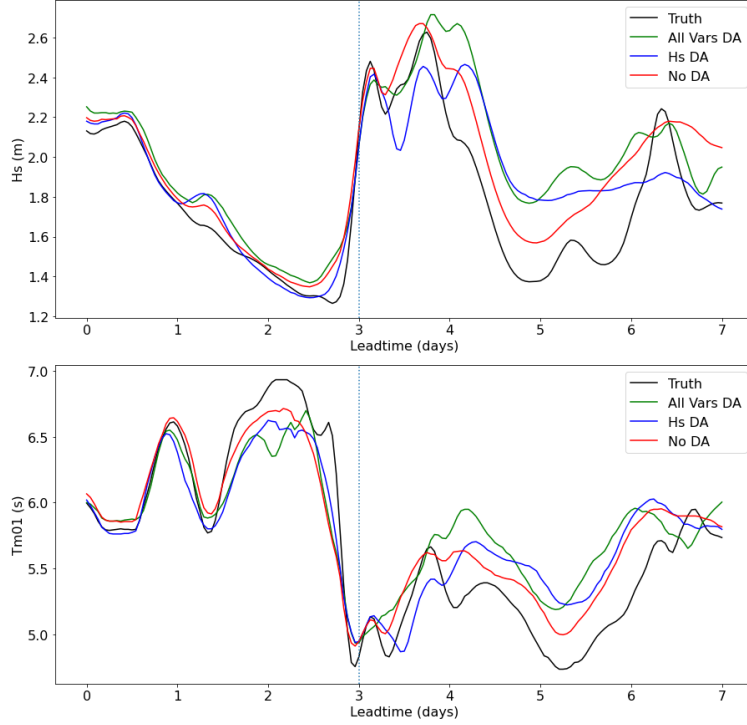


FIGURE 5.2: Timeseries of H_s (top) and T_{m01} (bottom) at $30^\circ\text{N } 80^\circ\text{W}$ from 12Z 09/03/2022 model run. No DA ensemble mean is shown in red, H_s DA ensemble is shown in blue, all vars DA ensemble mean is shown in green and truth is shown in black. Blue dotted line indicates the end of the assimilation window.

expect the increments to be small in this situation. We have seen that generally $\mathbf{P}^f$ is small at locations far from the buoy, with some exceptions noted. (and at times far from the assimilation window,)

From equation 2.15, the increments are a linear function of the innovations $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}_b$. We expect the increments and the innovations to have the same sign, so that the DA moves the analysis state closer to the truth than the forecast state.

Figure 5.3 shows the increments in H_s ensemble mean at 12Z 12/03/2022 taken from the 12Z 10/03/2022 forecast run. We see that all vars DA has larger increments overall, and this is especially true far from the buoy. The increments from both DA runs match the sign of the innovations somewhat well, but both have regions where the sign is incorrect. A notable example is the strong positive increment from all vars DA in the Northern GoM, where the innovation is slightly negative. The lower magnitude of increments in H_s DA means it undercorrects in some regions where the magnitude of innovations is large, such as just Northwest of the buoy in the Western GoM. All vars DA does a better job at correcting for this large innovation.

Figure 5.4 shows the same as 5.3 but for the increments and innovations in T_{m01} . We see again that all vars DA show larger increments overall, and here all vars DA does a better job of matching the sign of innovation across the wider domain. It captures the strip of negative innovations seen extending through the central GoM, which is missed by H_s DA. However just East of the buoy, H_s DA has positive increments matching the innovation, while all vars DA is strongly negative.

Overall, all vars DA has generally larger increments across the whole domain and

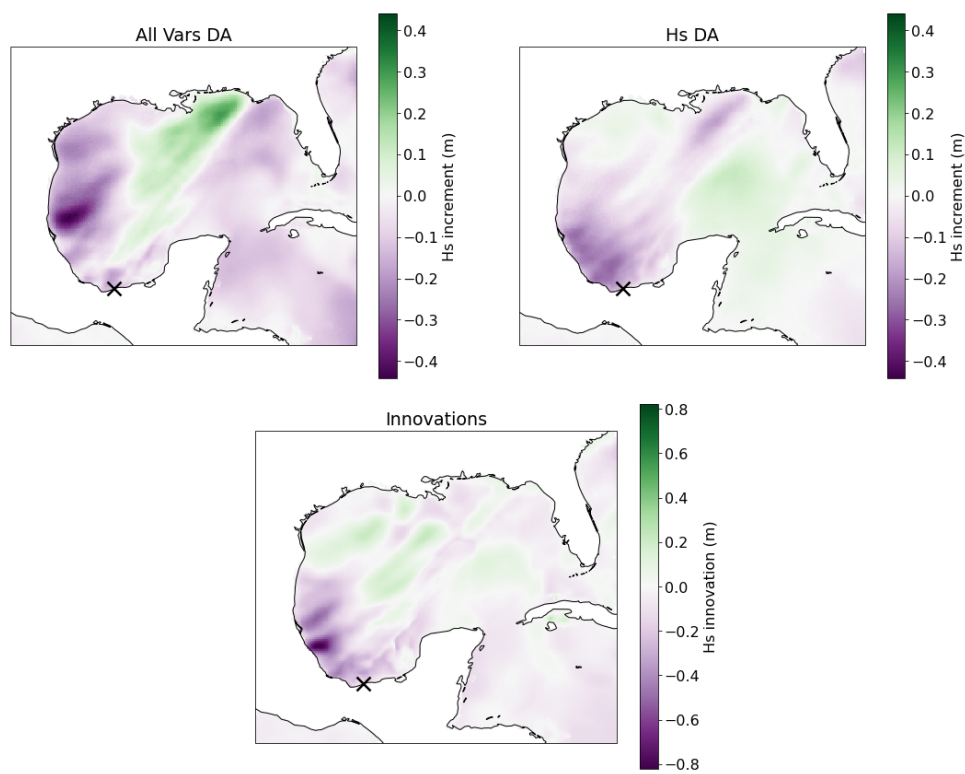


FIGURE 5.3: H_s ensemble mean increments at 12Z 12/03/2022 from all vars DA (top left) and H_s DA (top right), and H_s innovations from the same time (bottom). Taken from 12Z 10/03/2022 forecast run. The buoy location is marked with a cross.

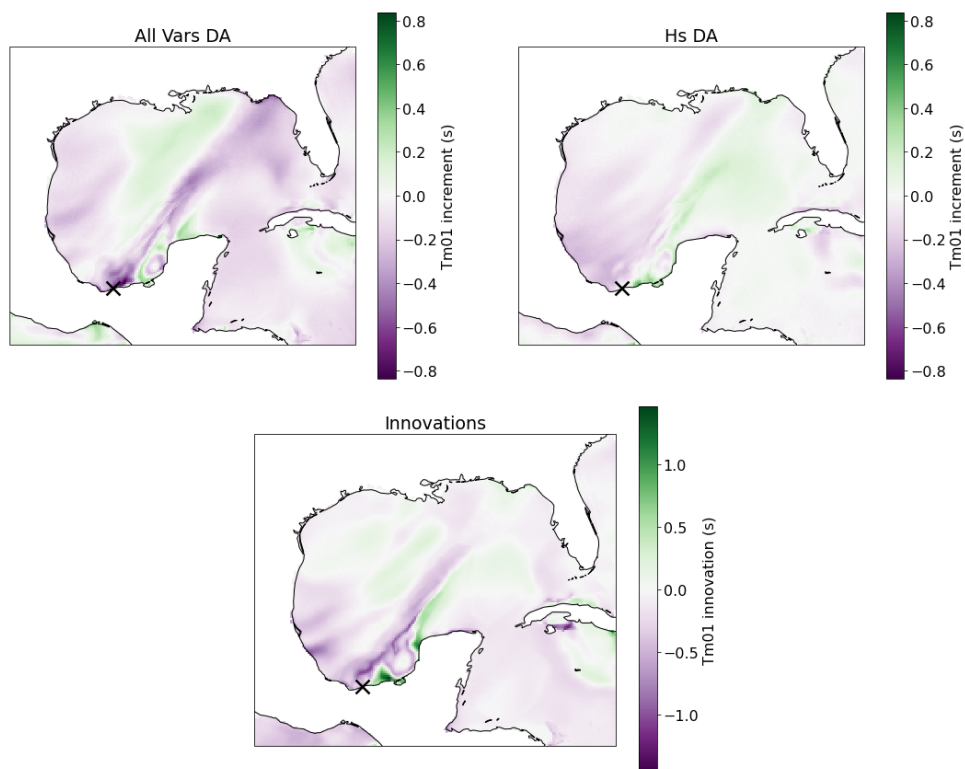


FIGURE 5.4: T_{m01} ensemble mean increments at 12Z 12/03/2022 from all vars DA (top left) and H_s DA (top right), and H_s innovations from the same time (bottom). Taken from 12Z 10/03/2022 forecast run. The buoy location is marked with a cross.

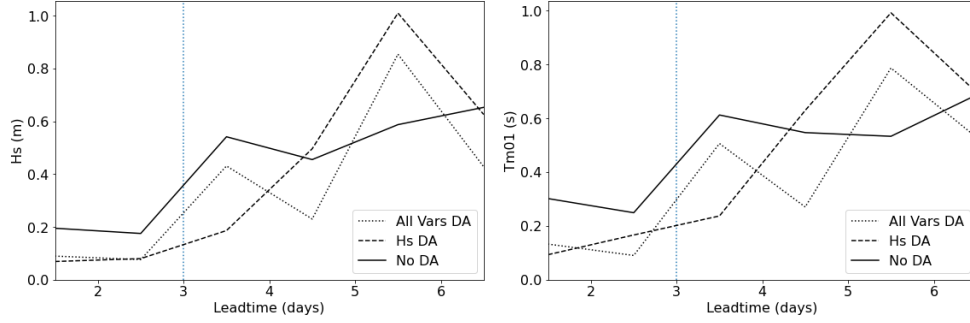


FIGURE 5.5: RMSE in H_s (left) and T_{m01} (right) ensemble mean at the buoy location as a function of leadtime. RMSE is calculated over a 24hr period. Results shown for no DA (solid line), H_s DA (dashed line) and all vars DA (black dotted line). Blue dotted line indicates the end of the assimilation window.

especially at locations far from the buoy. These large long distance increments probably arise from the long distance covariances seen in section 4.1. In that section, we concluded that some of the long distance covariances probably arise spuriously from sampling error, so we judge these increments to be somewhat physically unrealistic. For H_s , both DA runs have increments which mostly match the sign of innovations, and for T_{m01} , all vars DA does better than H_s DA.

5.2 RMSE

Figure 5.5 shows the RMSE of the ensemble mean from no DA, H_s DA and all vars DA. The mean is taken over a 24-hour period from 12Z 12/03/2022 to 12Z 13/03/2022. This period was chosen because it contains the highest waves in the forecast period, which are associated with higher errors. Forecasts of this period are available from multiple start dates, giving a range of leadtimes. Errors are plotted for H_s and T_{m01} . T_{m02} is not included as it has a very similar form to T_{m01} .

For H_s , we see that during the assimilation window both H_s DA and all vars DA show an improvement of $\sim 60\%$ over no DA. After the end of the window, the errors from all runs increase and at 3-4 days leadtime, H_s DA shows a much greater RMSE reduction than all vars DA. At 4-5 days, H_s DA has similar error to no DA and all vars DA shows an improvement of $\sim 50\%$. 6-7 is similar except the improvement from all vars DA is around 35 %. At 5-6 days leadtime, both H_s DA and all vars DA have higher RMSE than no DA. (why)

For T_{m01} the results are similar, with consistent improvement from DA within the assimilation window and more variable results afterwards. All vars DA still performs better than H_s DA with the exception of 1-2 days and 3-4 days leadtime. It is notable that H_s DA gives improvements in this case since no observations of T_{m01} are actually assimilated. This can be explained by the nonzero cross-covariance between H_s and T_{m01} (figure 4.6). One difference between the errors in T_{m01} and H_s and that H_s DA does slightly worse in predicting T_{m01} , when using the other two runs as a benchmark. This is unsurprising as we would expect assimilating H_s , T_{m01} and T_{m02} to give better predictions of T_{m01} than H_s alone.

Overall, the performance during the assimilation window is similar during the assimilation window, and after the window assimilating all variables gives better errors than

assimilating wave height alone.

Figure 5.6 shows the spatial distribution of RMSE in H_s ensemble mean over the period from 12Z 12/03/2022 to 12Z 13/03/2022. The middle panels show the RMSE from no DA, and the side panels show the RMSE reduction achieved by all vars DA i.e. $\text{RMSE}_{\text{No DA}} - \text{RMSE}_{\text{DA}}$. So when this quantity is positive, shown by blue regions on the plot, the RMSE with DA is smaller than without.

At 2-3 days leadtime, both H_s DA and all vars DA show significant improvements in the immediate vicinity of the buoy. H_s DA shows RMSE reduction in the western GoM that is weaker than the reduction near the buoy, and the eastern GoM has increased RMSE. Outside the GoM there is very little change. All vars DA has more regions within the GoM where the RMSE is increased, and the increase in the eastern GoM is larger than that shown by H_s DA. There is also a noticeable reduction outside the GoM.

At 4-5 days leadtime, all vars DA shows reduced RMSE in the western GoM and increased RMSE further from the buoy in the eastern GoM. The change in RMSE outside the GoM is smaller relative to that seen within the GoM, but the magnitude of RMSE change over the whole domain is generally larger than that seen at 2-3 days. This is probably due in part to the fact that the no DA RMSE is larger. H_s DA shows smaller RMSE changes overall than all vars DA, and there is no significant reduction near the buoy.

Figure 5.7 shows the spatial distribution of RMSE in T_{m01} ensemble mean over the period from 12Z 12/03/2022 to 12Z 13/03/2022. At 2-3 days leadtime, both H_s DA and all vars DA show improvements over no DA in the vicinity of the buoy with some increased errors in the eastern GoM and in the Atlantic. The general scale of error reduction/increase is the same in both cases. All vars DA also has regions of increased errors in the western GoM.

At 4-5 days leadtime, all vars DA shows improvement through most of the western GoM, with some increased errors again in the eastern GoM and in the Atlantic. H_s DA shows little change in RMSE through most of the domain, with the exception of a large increase in the northeastern GoM.

Overall, all vars DA does better than H_s DA at predicting both H_s and T_{m01} near the buoy after the assimilation window, while both perform similarly during the window. The improvements seen during the window are more widespread in H_s DA, while after the end of the window the opposite is true. All vars DA shows larger changes in errors in general, and particularly at locations away from the buoy, which is in line with the larger increments seen in all vars DA.

5.3 Correlation

5.4 Ensemble Spread

Ensemble spread is an important measure of uncertainty in ensemble forecasts. One aim of this project is to reduce uncertainty in the forecasts so that they may be used to inform decisions with greater confidence. Figure 5.8 shows timeseries of ensemble standard deviation in H_s and T_{m01} at the buoy location from two different model runs. Both H_s DA and all vars DA show a reduction in ensemble spread in both variables, as is expected from DA (section 2.2.3).

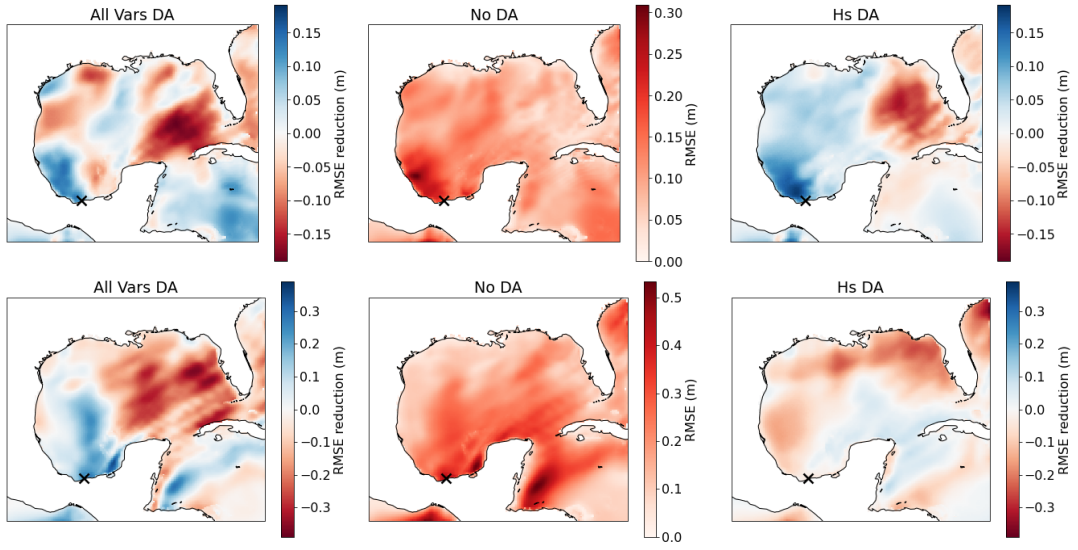


FIGURE 5.6: Maps of RMSE in H_s ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and H_s DA (right). The buoy location is marked with a cross.

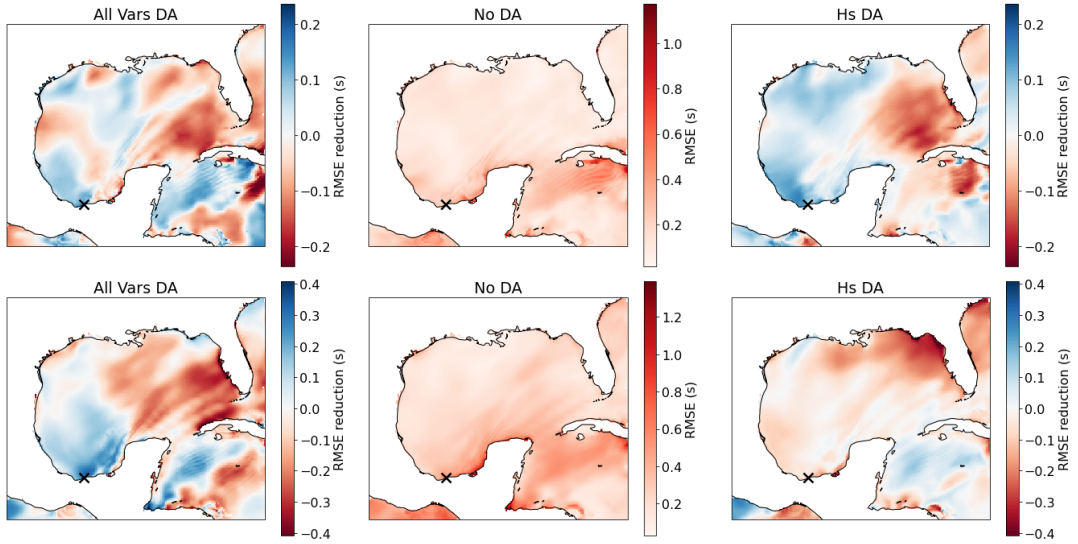


FIGURE 5.7: Maps of RMSE in T_{m01} ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and H_s DA (right). The buoy location is marked with a cross.

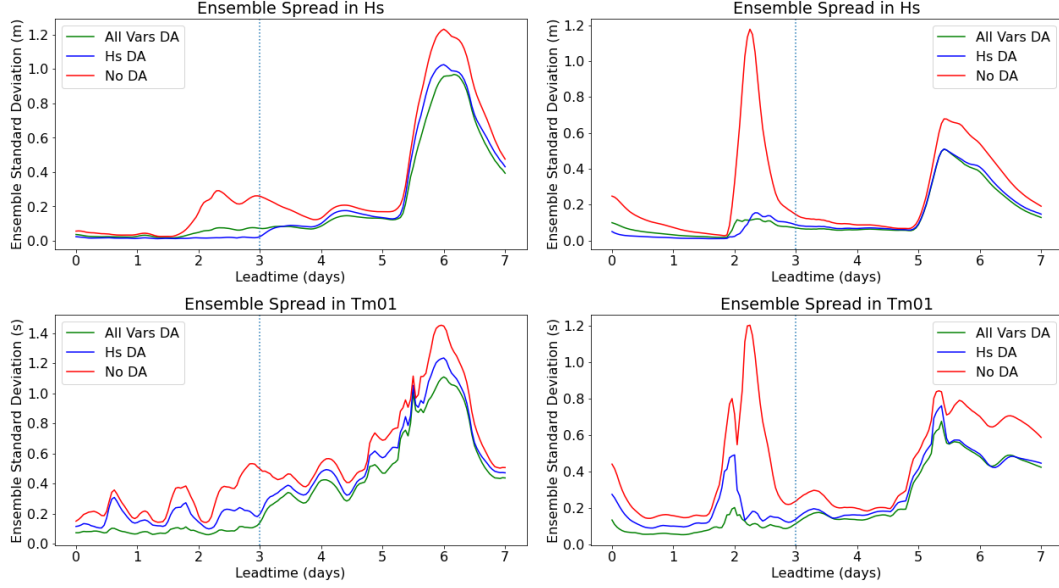


FIGURE 5.8: Timeseries of ensemble standard deviation in H_s (top) and T_{m01} (bottom) at the buoy location. Taken from 12Z 06/03/2022 (left) and 12Z 10/03/2022 (right) model runs.

All vars DA shows a greater reduction in T_{m01} ensemble spread overall. This is unsurprising since T_{m01} is not directly assimilated in H_s DA. This means that any increment to T_{m01} from H_s DA comes from cross covariances between H_s and T_{m01} , which we have seen are small (figure 4.6), so we expect the overall effect of H_s DA on T_{m01} to be smaller than the effect of all vars DA. In H_s ensemble spread, both runs perform similarly but H_s DA generally shows lower spread during the assimilation window and the opposite is true after the assimilation window. This is likely because the observational error scale in H_s was increased from 10% in H_s DA to 12% in all vars DA.

Figure 5.9 shows the spatial distribution of H_s ensemble spread at 12Z 11/03/2022, from 1 day and 5 days leadtime. At both leadtimes, All vars DA shows a greater reduction in ensemble spread than H_s DA across most of the domain. The only exception to this is the vicinity of the buoy at 1 day leadtime, when H_s DA shows a greater reduction. This is similar to what we saw in the timeseries, when H_s DA had greater spread reduction outside of the assimilation window. This reflects the fact that the entire 7-day forecast window is updated in one step by the DA system, so the dimensions of space and time within a forecast window are treated equivalently. As a result of this, is it not surprising to see parallels in the spatial and temporal distributions of ensemble spread. The greatest reductions in spread are where the spread of no DA is greatest.

Figure 5.10 shows the same as 5.9 but for T_{m01} . At 1 day leadtime, the pattern of reductions from both all vars DA and H_s DA closely match the pattern of standard deviation from no DA, and the reductions from all vars DA are stronger overall. At 5 days leadtime, the reduction from all vars DA generally matches the pattern of standard deviation from no DA, with a peak concentrated on the western side of the Yucatan peninsula. This peak is also present in H_s DA, which shows little reduction in spread elsewhere.

Overall, all vars DA shows greater reduction in ensemble spread at most times and

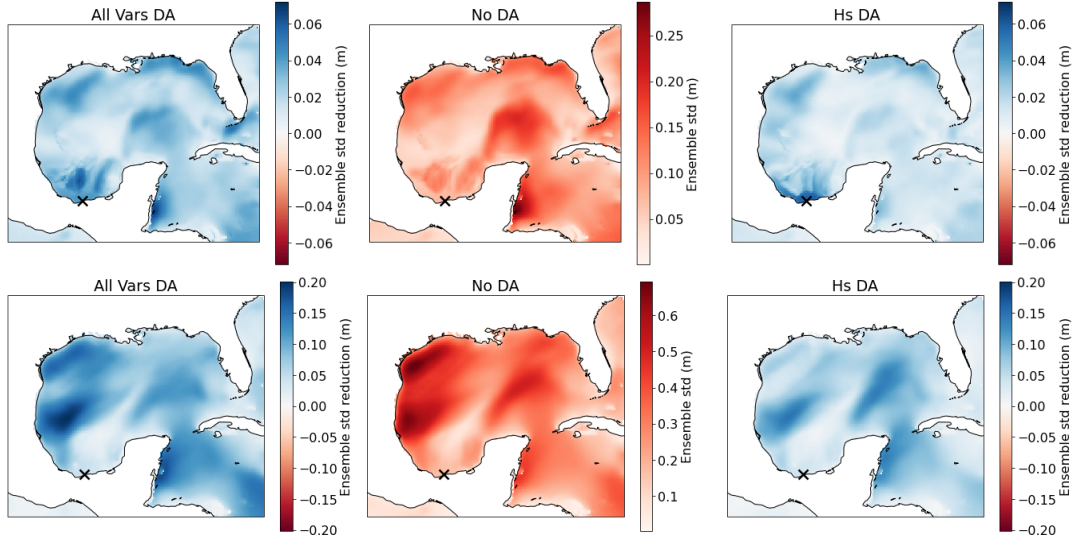


FIGURE 5.9: Ensemble standard deviation in H_s at 12Z 11/03/2022, taken from 12Z 06/03/2022 (top) and 12Z 10/03/2022 (bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation reduction from all vars DA (left) and H_s DA (right).

locations, except near the buoy during the assimilation window. The reductions in spread are greatest where the no DA ensemble spread is greatest, which is good as this reduces the maximum uncertainty in the forecast.

It is not desirable to focus solely on reducing the ensemble spread, as this may lead to the ensemble being overconfident. As described in section 2.2.3, it is detrimental to the forecast to have ensemble spread smaller than the forecast errors as this can lead to too much confidence being placed in an inaccurate forecast. To test whether this problem has arisen, we plot the ensemble standard deviation against the RMSE.

Figure 5.11 shows a scatter plot of ensemble mean against RMSE at the buoy location. RMSE is calculated from the ensemble mean and averaged over a 6 hour rolling window in leadtime. Data from all forecast windows from H_s DA and all vars DA is included, with H_s DA represented by circles and all vars DA represented by triangles. The leadtime of each data point within its forecast window is shown by its colour. Lines of best fit are calculated for both H_s DA and all vars DA, and plotted along with the 1:1 line.

We see from the lines of best fit that both DA runs have RMSE generally below the ensemble spread, with all vars DA being lower than H_s DA overall. Data points above the 1:1 line correspond to overconfident ensembles, so for both variables from both DA runs, the number of data points above the line was counted and shown in table 5.1. The equivalent counts from no DA are also included. The total number of data points in each run is 984, so we see that all vars DA gives overconfident H_s ensembles about 22% of the time and overconfident T_{m01} ensembles about 30% of the time. H_s DA gives overconfident H_s ensembles about 30% of the time and overconfident T_{m01} ensembles about 25% of the time. The colours on the plot show that most of the occasions where RMSE is significantly more than spread come late in the forecast window (leadtime > 4 days) with few exceptions.

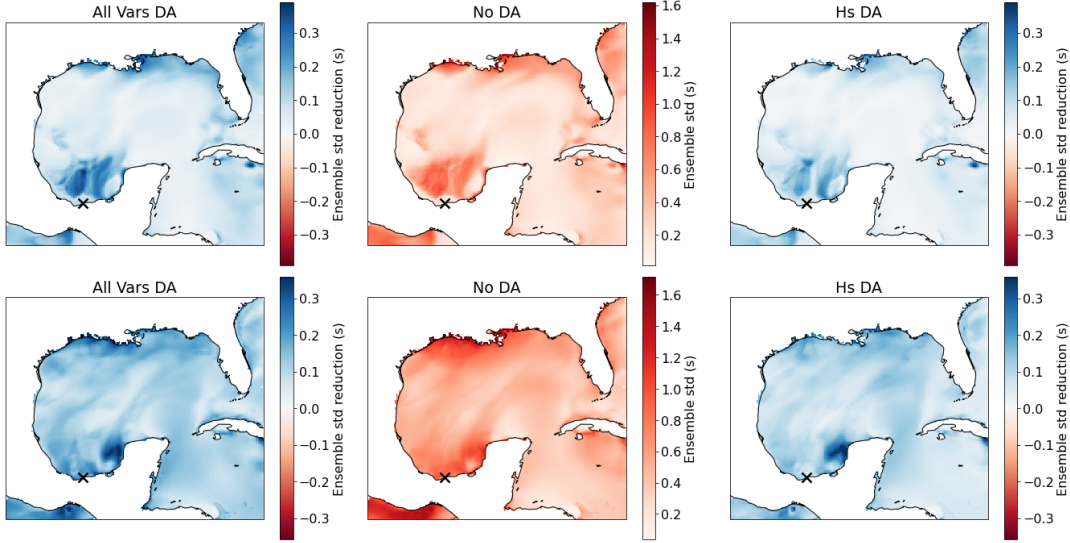


FIGURE 5.10: Ensemble standard deviation in T_{m01} at 12Z 11/03/2022, taken from 12Z 06/03/2022 (top) and 12Z 10/03/2022 (bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation reduction from all vars DA (left) and H_s DA (right).

No RMSE Threshold			With RMSE Threshold		
	H_s freq.	T_{m01} freq.		H_s freq.	T_{m01} freq.
No DA	21	19	No DA	8	18
H_s DA	291	250	H_s DA	115	214
All Vars DA	212	293	All Vars DA	100	135

TABLE 5.1: Left: Frequency of overconfident H_s and T_{m01} ensembles from no DA, H_s DA and all vars DA. Right: The same but the ensemble must meet a minimum RMSE threshold of 0.1m for H_s and 0.2s for T_{m01} to be counted.

So H_s DA gives more overconfident H_s ensembles and all vars DA gives more overconfident T_{m01} ensembles, with H_s DA giving more overconfident ensembles overall. The high prevalence of overconfident T_{m01} ensembles from all vars DA lines up with the decreased spread in T_{m01} from all vars DA seen in figure 5.8, and may be due to the increased reduction in ensemble spread from multiple variables being assimilated.

Overconfident ensembles are most important to avoid when the error is large, since larger forecast errors present greater risk for incorrect decision making. So we repeat the count of the overconfident ensembles and apply a minimum threshold that the RMSE must meet in order for the ensemble to be counted. These thresholds are 0.1 m for H_s and 0.2 s for T_{m01} , and were set at values that we judge to be acceptable levels of error in each variable. After applying these thresholds, the frequency of overconfident ensembles reduced by at least 50% in all cases except the T_{m01} ensembles from H_s DA. This reflects the fact that the errors in T_{m01} from H_s DA are large. This now gives overconfidence prevalences of 10-20% in most cases.

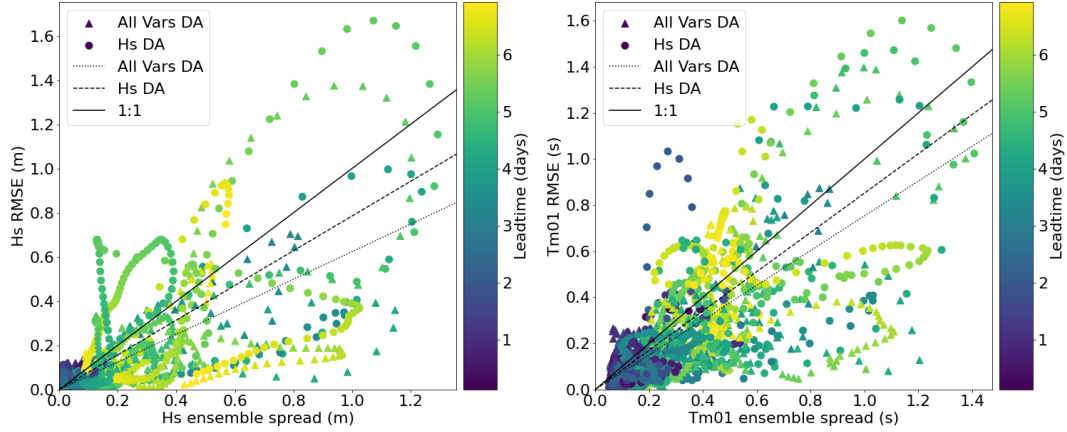


FIGURE 5.11: Left: Scatter plot of H_s RMSE against H_s ensemble standard deviation at the buoy location. All vars DA shown with triangles, H_s DA shown with circles and leadtime shown by colour. Lines of best fit shown for all vars DA (dotted) and H_s DA (dashed) as well as 1:1 line (solid). Right: the same for RMSE and ensemble spread in T_{m01} .

5.5 Errors in Maximum H_s

H_s maxima are of particular interest in this project as they present the most danger to offshore operations. In this section we examine the effect of DA on the errors in the timing and magnitude of the H_s maximum on 12/03/2022. Figure 5.2 shows scatter plots of the timing and magnitude from no DA, H_s DA and all vars DA, relative to the truth. The maximum is selected from the period spanning 12Z 12/03/2022 to 12Z 13/03/2022. The left plot is generated from the model run starting 12Z 10/03/2022 and the right plot is generated from the model run starting 12Z 08/03/2022, giving leadtime ranges of 2-3 days and 4-5 days respectively. Table 5.2 shows the ensemble RMSE in timing and magnitude from each ensemble for both runs.

Note on the scatter plots that the points are arranged discretely at integer numbers of hours: this is simply because the forecast data used has a time resolution of one hour. This could reduce the maximum wave height detected if the true maximum lies in the middle of two timesteps. One way around this may be to fit a quadratic polynomial to the model data points either side of the peak and find the maximum value of this polynomial, but this was not explored here. The forecasters at Fugro work with hourly forecast data without any interpolation applied, and we assume they are aware that this can lead to inaccurate maximum values being sampled.

With a leadtime of 2-3 days, we see a large improvement in forecasts of timing and magnitude from both DA runs. From the scatter plot, the ensemble spread is considerably tighter around the true value, and the RMSE in both quantities is reduced. All vars DA does slightly better than H_s DA in predicting the timing of the maximum, and vice versa for the magnitude, but the differences between the two are small compared to the difference of both with no DA.

After 4-5 days, the errors are larger in all quantities but the magnitude from the no DA run. The differences between the two DA runs is larger, as H_s DA has a similar error in timing to no DA, but shows a small improvement in the magnitude of 13%. Conversely, all vars DA has similar error in magnitude to no DA, but shows an improvement in timing of 35%.

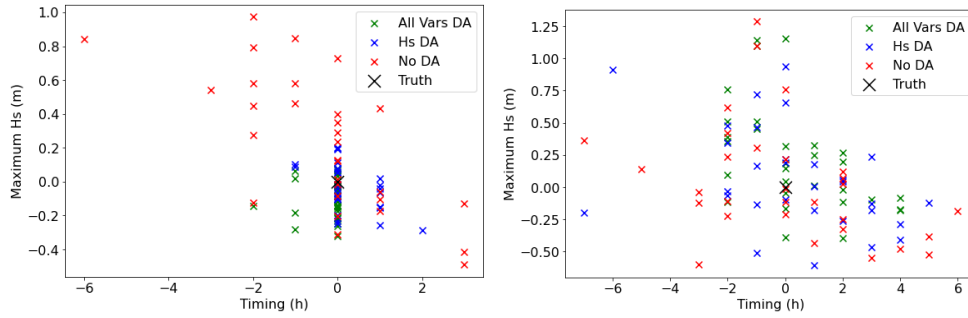


FIGURE 5.12: Scatter plots of timing and magnitude of maximum H_s at buoy location from period 12Z 12/03/2022 to 12Z 13/03/2022. Generated from model run starting 12Z 10/03/2022 (left) and 12Z 08/03/2022 (right). Truth is shown in black, and the no DA, H_s DA and all vars DA ensemble members are shown in red, blue and green respectively.

Leadtime 2-3 days

	Timing (h)	Magnitude (m)
No DA	1.9	0.46
H_s DA	0.66	0.14
All Vars DA	0.51	0.15

Leadtime 4-5 days

	Timing (h)	Magnitude (m)
No DA	2.9	0.45
H_s DA	2.7	0.40
All Vars DA	1.9	0.46

TABLE 5.2: RMSE of magnitude and timing of the H_s maximum from the ensembles plotted in figure .

Overall, H_s DA and all vars DA perform similarly to each other during the assimilation window, and both significantly outperform no DA as we expect during this time. As seen in other aspects, their performance differs more after the assimilation window, with H_s DA providing better forecasts of the magnitude of the peak and all vars DA providing better forecasts of its timing.

Chapter 6

Conclusions

6.1 Summary

We assimilated observations of H_s , T_{m01} and T_{m02} into Fugro’s wave forecasts in the GoM region. To do this, we made use of the update step from the ETKF method to correct an entire 7-day forecast run in one go. This differs from conventional ETKF implementations, in which simultaneous observations are assimilated together and then the forecast model is used to step the state forward to the next observation time.

Investigating the background errors, we found that the ensemble variance in H_s is a strong function of H_s itself, and it peaks just before the peak in H_s . A similar relationship was found for T_{m01} . We saw that the forecast error covariances are often strongest near the buoy location, and longer distance covariances can arise physically due to winds and travelling waves and also spuriously due to sampling error.

We found that observation errors are a strong linear function of the variable being observed, particularly in H_s . This motivated us to set the observation error standard deviation in the DA system to a linear scale of the observations. Since the scaling factor for the periods was small, we implemented a constant minimum error threshold for these variables to ensure that the error would not become too small and negatively impact DA performance.

We compared the results of assimilating H_s alone (H_s DA) to assimilating all three variables (all vars DA), and found that both approaches, but particularly all vars DA, produced somewhat unrealistic increments with large changes to the state far away from the observation location. Comparing our results to Houghton et al. (2023) (figure 1.3 and figure 5.3), we see that their LETKF method produced more realistic increments, which captured the flow dependency of the background covariances while keeping the effects of the observations confined to the vicinity of the observation location.

We found reductions in RMSE of up to 50% in both variables, which extended up to 0.5 days beyond the end of the assimilation window in the case of H_s DA, and 1.5 days beyond in the case all vars DA. Both cases produced very similar RMSE reductions during the assimilation window, and all vars DA had consistently smaller errors than H_s DA at >1 day after the end of assimilation. At 2-3 days after assimilation, both DA runs had larger errors than the no DA case (and we were unable to find an explanation for this).

These improvements in RMSE are less consistent and slightly shorter lived than those obtained in Houghton et al., 2022, 2023 (figures 1.3, 1.4). This reflects the two major limitations identified in section 1.2, namely the limited number of observations

available to us and the fact that we do not have access to the forecast model, which means that we must update an entire forecast run in one update step and not use cycling. Our results agree with Houghton et al., 2022 in that assimilating multiple variables (spectral DA) performed better than assimilating wave height alone in most cases. One difference between the results of that study and this is that Houghton et al., 2022 found H_s DA and spectral DA to be virtually identical in predicting H_s , while spectral DA was better for predicting other variables. We found that all vars DA did better than H_s DA in all the variables tested, and this may reflect our choice to assimilate a selection of bulk variables rather than the full spectrum.

The RMSE improvements we found are more similar to those found in Caires, Kim, and Groeneweg, 2018, which likely reflects the smaller number of observations used in that study. The fact that we achieved similar results to Caires, Kim, and Groeneweg, 2018 despite not being able to run the forecast model may indicate that the limited number of observations is the more important limitation out of the two. It may also be due to other differences in methodology between this study and that, such as a different DA method, compensating for the lack of a forecast model in this study.

Looking at the wider domain, we found that H_s DA showed more consistent and widespread RMSE reductions during the assimilation window, and the opposite is true after the window.

We found that the ensemble is reduced by both all vars DA and H_s DA. For H_s forecasts, H_s DA showed greater reductions in spread near the buoy during the assimilation window and all vars DA showed greater reductions at other locations and times. We found that post DA, there was some degradation in the suitability of ensemble spread as a measure of forecast error. There were many more instances of what we call ‘overconfident’ ensembles, where there is a large error which the ensemble spread does not capture. We found that the prevalence of overconfident ensemble went from 1-2% before DA to around 20-30% afterwards.

This may pose a risk if the forecasts are used to inform decision making in offshore operations. This aspect of performance could be improved with inflation and/or further tuning of observation error scales, and other improvements to the DA method such as the introduction of cycling could reduce error overall, which would also help in this regard.

DA was found to reduce the errors in forecasts of the H_s maximum associated with the cold front on 12/03/2022. With a 2-3 day leadtime (so the maximum was during the assimilation window), both H_s DA and all vars DA showed improvements showed improvements of around 65% in forecasts of both the timing and magnitude of the H_s maximum. At 4-5 days leadtime, when the maximum was outside of the assimilation window, all vars DA showed improvement in forecasting the timing of the peak, while H_s DA did better forecasting its magnitude.

6.2 Limitations and Future Work

As noted before, two major limitations of this study are the small number of observations available and the lack of access to a forecast model which makes cycling impossible. The first of these limitations could be mitigated by augmenting the observations used here with other wave observations in the GoM. NOAA maintains a public database of up to date observations from many fixed buoys around the world. Several of these buoys are located in the GoM and measure the same bulk quantities

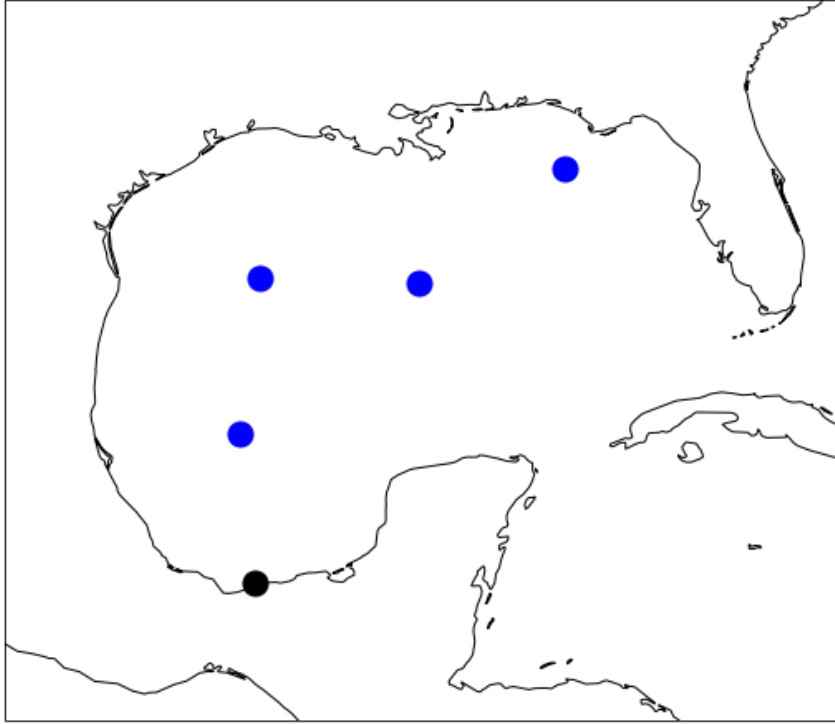


FIGURE 6.1: Locations of buoys with observations of relevant wave variables. The buoy used in this project is marked in black, and other buoys are marked in blue

assimilated in this project. A map showing the locations of a selection of these buoys is shown in figure 6.1.

As demonstrated in section 1.2, DA tends to do better when it has more observations to work with, and so including data from these buoys may improve the DA performance. It is also worth restating here that the twin experiment methodology used here excluded the possibility of correcting any biases present in the forecast data, so future work assimilating real observations would be useful to investigate the effectiveness of DA for this purpose.

As well as having limited observations, we also had limited access to forecast data, with only seven overlapping week-long forecast windows available. This raises questions over the robustness of the DA results, and so it would be useful for future studies to include more forecast data.

The lack of a forecast model will not be an issue in any future operational DA implementation undertaken by Fugro. There are many ways in which the method used in this study could be modified to include cycling, and perhaps one of the simplest is to assimilate the observations as they are available in a more conventional ETKF setup. This would reduce the dimension of the state space since the state vector would only contain values from a single timestep. Observations could be assimilated every hour, or some observations could be skipped so that the DA could be performed less frequently to reduce computational cost. We have seen in this study that the ETKF reduces the ensemble spread in comparison to the forecast error, so this proposed method would be susceptible to filter divergence and some inflation or other preventative measure would likely be necessary to mitigate this.

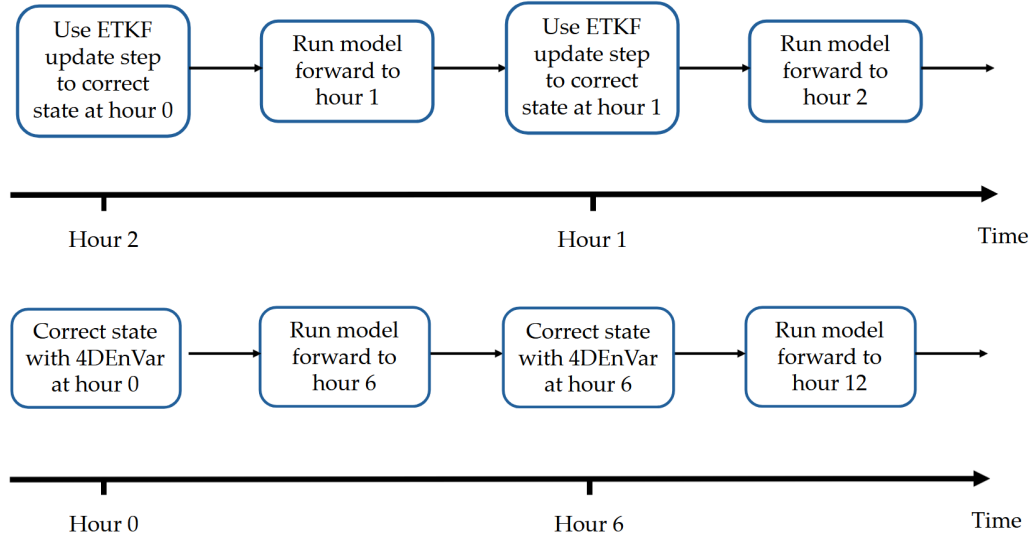


FIGURE 6.2: Schematics of potential DA methods which incorporate cycling. Top: Observations assimilated hourly using a conventional ETKF filter method. Bottom: Observations assimilated every 6 hours using the 4DEnVar smoother method.

Another possible approach would be to use a smoother method such as 4DEnVar. A smoother method simultaneously assimilates observations from an entire window, rather than assimilating observations from each observation time one by one as in a filter method. This would allow assimilation to be done less frequently while still assimilating all available observations, and is less prone to filter divergence. However, this method would require the augmentation of the state vector to include values from multiple timesteps, and would require minimisation of the cost function using a method such as the conjugate gradient method (Nocedal and Wright, 2006) which is typically computationally expensive. Further research should be done to determine the best approach. Schematics of both methods outlined here are shown in figure 6.2.

Since we have seen that the ensemble spread is reduced too much by the DA method used in this project, future studies into this topic should include some kind of countermeasure against this effect. One possible countermeasure which has been discussed is inflation applied prior to the update step, which involves multiplying $\mathbf{P}^f$ by some predefined factor $\alpha > 1$ before updating the model states. It is also possible to apply inflation during the analysis step by multiplying $\mathbf{P}^a$ by α prior to generating the analysis ensemble.

Other potential methods for artificially increasing the ensemble are adding noise to the forecast ensemble prior to calculating $\mathbf{P}^f$ (Anderson and Anderson, 1999), or the method of relaxation from Whitaker and Hamill, 2012, in which the analysis ensemble is modified to bring its standard deviation closer to that of the forecast ensemble while leaving its mean unaffected.

Another problem encountered in this study is the large covariances and increments seen far away from the observation location. As discussed, this is often physically unrealistic, with spurious long distance covariances in $\mathbf{P}^f$ arising from sampling error and causing unrealistic large long-distance increments (Hunt, Kostelich, and Szunyogh, 2007). This may be improved with localisation, but since at least some of the

long-distance covariances seen appeared to be genuine, we did not explore this in this study and left it instead for future work. Further investigation into the physical mechanisms behind the covariances would also be helpful here to inform the implementation of localisation.

Appendix A

4DEnVar Method

Another way to assimilate observations from multiple times is the 4DVar method. This method uses a form of the cost function from equation 2.14, augmented to include multiple observations. Let $t = 0, 1, \dots, T$ be the observation times, then we define the model operator $\mathcal{M}_{0 \rightarrow t}$ such that the state predicted by the model at time t is related to the state at time 0 by $\mathbf{x}_t = \mathcal{M}_{0 \rightarrow t} \mathbf{x}_0$. The 4DVar cost function then has the form

$$J(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \mathbf{H}_t \mathcal{M}_{0 \rightarrow t} \mathbf{x}_0\|_{\mathbf{R}_t^{-1}}^2 \quad (\text{A.1})$$

or equivalently

$$J(\delta_0) = \frac{1}{2} \|\delta_0\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \mathbf{H}_t \mathcal{M}_{0 \rightarrow t} (\mathbf{x}_b + \delta_0)\|_{\mathbf{R}_t^{-1}}^2 \quad (\text{A.2})$$

Where $\delta_0 = \mathbf{x}_0 - \mathbf{x}_b$. In this cost function, the observations $\mathbf{y}_t$ and matrices $\mathbf{H}_t, \mathbf{R}_t$ are valid at time t . Note here that $\mathbf{x}_0$ is a non-augmented state vector containing values only from time 0.

The above equations are valid for a single deterministic forecast. If an ensemble is available 4DEnVar is a variant of 4DVar in which a control variable transform is used to write $\mathbf{x}_0$ as a linear combination of the ensemble perturbations:

$$\begin{aligned} \mathbf{X}'_b &= \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_1^f - \bar{\mathbf{x}}^f, \mathbf{x}_2^f - \bar{\mathbf{x}}^f, \dots, \mathbf{x}_N^f - \bar{\mathbf{x}}^f \right) \\ \mathbf{x}_0 &= \bar{\mathbf{x}}_b + \mathbf{X}'_b \mathbf{w}_0 \end{aligned}$$

Where $\bar{\mathbf{x}}^f$ and $\mathbf{X}^f$ are defined in equation 2.17. The cost function can then be rewritten in terms of χ and simplified by linearising $\mathbf{M}_{0 \rightarrow t}$ around $\bar{\mathbf{x}}^f$:

$$J(\chi) = \frac{1}{2} \|\chi\|^2 + \frac{1}{2} \sum_{t=0}^T \left\| \mathbf{y}_t - \mathbf{H}_t \mathbf{M}_{0 \rightarrow t} \left(\bar{\mathbf{x}}^f + \mathbf{X}^f \chi \right) \right\|_{\mathbf{R}_t^{-1}}^2 \quad (\text{A.3})$$

This cost function has been simplified by taking a background error covariance matrix of the form $\mathbf{B} = \mathbf{X}^f \mathbf{X}^{f\top}$, which reduces to the identity under the control variable

transform. Note that this is identical to the forecast error covariance matrix used in ETKF. This cost function has gradient

$$\nabla J(\chi) = \chi + \sum_{t=0}^T \left(\hat{\mathbf{H}}_t \mathbf{X}'_b \right)^\top \mathbf{R}_t^{-1} \left(\hat{\mathbf{h}}_t \mathbf{X}'_b \chi + \hat{\mathbf{H}}_t \bar{\mathbf{x}}_b - \mathbf{y}_t \right)$$

A major advantage of this formulation is that we can make the approximation

$$\mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right) \approx \mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} \right) - \overline{\mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b \right)}$$

Since $\mathbf{H}_t \mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right)$ is the n^{th} column of $\hat{\mathbf{H}}_t \mathbf{X}'_b$, this approximation allows us to compute the cost function and its gradient without a tangent linear or adjoint model, which is a computationally expensive aspect of most variational data assimilation schemes (Bannister, 2017).

Bibliography

- Anderson, Jeffrey L and Stephen L Anderson (1999). “A Monte Carlo Implementation of the Nonlinear Filtering Problem to Produce Ensemble Assimilations and Forecasts”. In: *Monthly Weather Review* 127 (12), pp. 2741–2758.
- Appendini, Christian M. et al. (Feb. 2014). “Wave Climate and Trends for the Gulf of Mexico: A 30-Yr Wave Hindcast”. In: *Journal of Climate* 27 (4), pp. 1619–1632. ISSN: 0894-8755. DOI: [10.1175/JCLI-D-13-00206.1](https://doi.org/10.1175/JCLI-D-13-00206.1). URL: <https://journals.ametsoc.org/view/journals/clim/27/4/jcli-d-13-00206.1.xml>.
- Asch, Mark, Marc Bocquet, and Maëlle Nodet (2016). *Data Assimilation: Methods, Algorithms and Applications*. Society for Industrial and Applied Mathematics. ISBN: 978-1-611974-53-9.
- Bannister, R. N. (2017). “A review of operational methods of variational and ensemble-variational data assimilation”. In: *Quarterly Journal of the Royal Meteorological Society* 143 (703), pp. 607–633. ISSN: 1477-870X. DOI: [10.1002/QJ.2982](https://doi.org/10.1002/QJ.2982).
- Bretherton, F P and C J R Garrett (Jan. 1968). “Wavetrains in inhomogeneous moving media”. In: *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences* 302 (1471), pp. 529–554. ISSN: 0080-4630. DOI: [10.1098/RSPA.1968.0034](https://doi.org/10.1098/RSPA.1968.0034). URL: <https://royalsocietypublishing.org/>.
- Caires, Sofia, Jinah Kim, and Jacco Groeneweg (Nov. 2018). “Korean East Coast wave predictions by means of ensemble Kalman filter data assimilation”. In: *Ocean Dynamics* 68 (11), pp. 1571–1592. ISSN: 16167228. DOI: [10.1007/S10236-018-1214-0](https://doi.org/10.1007/S10236-018-1214-0). URL: <https://link.springer.com/article/10.1007/s10236-018-1214-0>.
- Chen, Yumeng, Lars Nerger, and Amos S. Lawless (2024). “EGUsphere - A Python interface to the Fortran-based Parallel Data Assimilation Framework: pyPDAF v1.0.0”. Preprint available at URL <https://egusphere.copernicus.org/preprints/2024/egusphere-2024-1078/#discussion>.
- Desroziers, G. et al. (Jan. 2006). “Diagnosis of observation, background and analysis-error statistics in observation space”. In: *Quarterly Journal of the Royal Meteorological Society* 131 (613), pp. 3385–3396. ISSN: 00359009. DOI: [10.1256/qj.05.108](https://doi.org/10.1256/qj.05.108).
- Gene H. Golub, Charles F. van Loan (1996). *Matrix Computations*. third. The John Hopkins University Press. ISBN: 0-8018-5414-8.
- Holthuijsen, Leo H. (2007). *Waves in Oceanic and Coastal Waters*. Cambridge University Press. ISBN: 978-0-521-12995-4.
- Houghton, Isabel A. et al. (Aug. 2022). “Operational Assimilation of Spectral Wave Data From the Sofar Spotter Network”. In: *Geophysical Research Letters* 49 (15), e2022GL098973. ISSN: 1944-8007. DOI: [10.1029/2022GL098973](https://doi.org/10.1029/2022GL098973).
- Houghton, Isabel A. et al. (June 2023). “Ensemble-based data assimilation of significant wave height from Sofar Spotters and satellite altimeters with a global operational wave model”. In: *Ocean Modelling* 183, p. 102200. ISSN: 1463-5003. DOI: [10.1016/J.OCEMOD.2023.102200](https://doi.org/10.1016/J.OCEMOD.2023.102200).
- Hunt, Brian R., Eric J. Kostelich, and Istvan Szunyogh (June 2007). “Efficient data assimilation for spatiotemporal chaos: A local ensemble transform Kalman filter”.

- In: *Physica D: Nonlinear Phenomena* 230 (1-2), pp. 112–126. ISSN: 0167-2789. DOI: [10.1016/J.PHYSD.2006.11.008](https://doi.org/10.1016/J.PHYSD.2006.11.008).
- Janssen, Peter (2004). *The Interaction of Ocean Waves and Wind*. Cambridge University Press. ISBN: 0-521-46540-0.
- Jazwinski, Andrew H. (1970). *Stochastic Processes and Filtering Theory*. Academic Press. ISBN: 978-0-486-46274-5.
- Lefèvre, J.-M. and L. Aouf (June 2012). “Latest developments in Wave in Data Assimilation”. eng. In: *ECMWF Workshop on Ocean Waves*.
- Miles, John W. (1957). “On the generation of surface waves by shear flows”. In: *Journal of Fluid Mechanics* 3.2, 185–204. DOI: [10.1017/S0022112057000567](https://doi.org/10.1017/S0022112057000567).
- Nocedal, Jorge and Stephen J. Wright (2006). *Numerical Optimization*. Second. Springer Science and Business Media. ISBN: 0-387-30303-0.
- Phillips, O. M. (1957). “On the generation of waves by turbulent wind”. In: *Journal of Fluid Mechanics* 2.5, 417–445. DOI: [10.1017/S0022112057000233](https://doi.org/10.1017/S0022112057000233).
- Vallis, Geoffrey K. (2017). *Atmospheric and Oceanic Fluid Dynamics*. Second. Cambridge University Press. ISBN: 987-1-107-06550-5.
- Whitaker, Jeffrey S. and Thomas M. Hamill (Sept. 2012). “Evaluating Methods to Account for System Errors in Ensemble Data Assimilation”. In: *Monthly Weather Review* 140 (9), pp. 3078–3089. DOI: [10.1175/MWR-D-11-00276.1](https://doi.org/10.1175/MWR-D-11-00276.1).
- Wu, Mengning and Zhen Gao (Sept. 2021). “Methodology for developing a response-based correction factor (alpha-factor) for allowable sea state assessment of marine operations considering weather forecast uncertainty”. In: *Marine Structures* 79, p. 103050. ISSN: 0951-8339. DOI: [10.1016/J.MARSTRUC.2021.103050](https://doi.org/10.1016/J.MARSTRUC.2021.103050).
- WW3DG, The WAVEWATCH III Development Group (2019). *User manual and system documentation of WAVEWATCH III version 6.07*. Tech. rep. 333. NOAA/NWS/NCEP/MMAB, College Park, MD, USA.